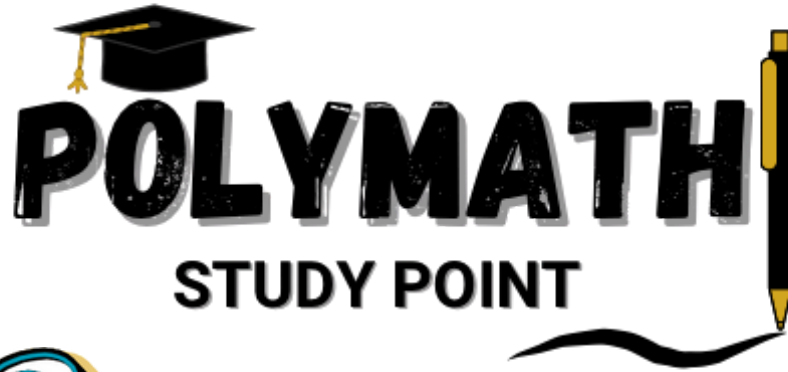


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Conquer Purvalokan

Revising The Past To Secure Future

Geography

FOR UPSC/UPPSC/BPSC/UPSSSC & OTHER COMPETITIVE EXAMINATIONS



INDIAN GEOGRAPHY

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INDIAN GEOGRAPHY

AREA & LATITUDINAL EXPANSION

Top 10 largest countries in the world (In area)



AREA OF INDIA

- **India's rank by area: 7th largest** in the world
 - Larger countries: **Russia, Canada, USA, China, Brazil, Australia**
- **Total area: 3.287 million sq. km** (approx. 2.4% of world's land surface)
- **Population (2011 Census): 17.5% of world population** lives in India
- **States & UTs: 28 States + 8 Union Territories**
- **Villages:**
 - 2001 Census – **6,38,588 villages**
 - 2011 Census – **6,40,932 villages**
- **Territorial waters:** extend up to **12 nautical miles (22.2 km)** from baseline

FACTS

- India is **7th largest** (not 5th or 6th).
- Accounts for **2.4% land area** but **17%+ population** of the world.
- **Dong (Arunachal Pradesh)** – first place in India to get sunrise.
- **Kanyakumari (8°4'N)** – southernmost point of mainland India.

- **Latitudinal extent: 8°4' N – 37°6' N**
- **Longitudinal extent: 68°7' E – 97°25' E**
- **North–South length: 3214 km**
- **East–West width: 2933 km**
- **Latitudinal spread: ~30°**
 - Each degree ≈ **111 km**
- **Longitude & Time difference:**
 - **1° longitude = 4 minutes** difference in local time
 - Gujarat (west) to Arunachal Pradesh (east) ≈ **29° longitude → ~2 hrs time lag**

LATITUDE BASICS

- **Equator (0° latitude)** divides Earth into **two hemispheres**.
- **Poles:** 90° N (North Pole), 90° S (South Pole).
- **Parallels of Latitude:** 179 in total (89 North + 89 South + Equator).

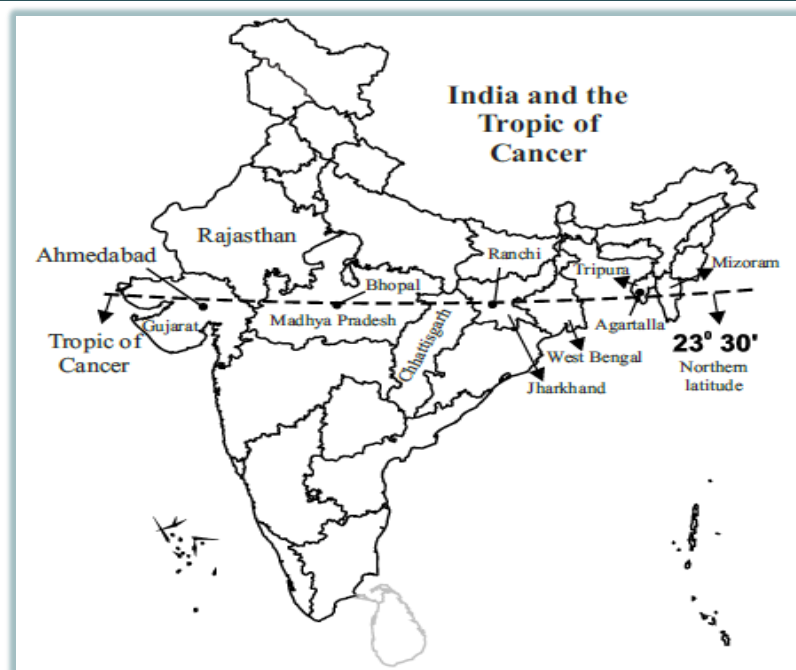
LONGITUDE BASICS

- **Prime Meridian (0° longitude):** passes through **Greenwich (London)**.
- **International Date Line (180° longitude).**
- **India** lies completely in the **Northern & Eastern Hemisphere**.

FACTS

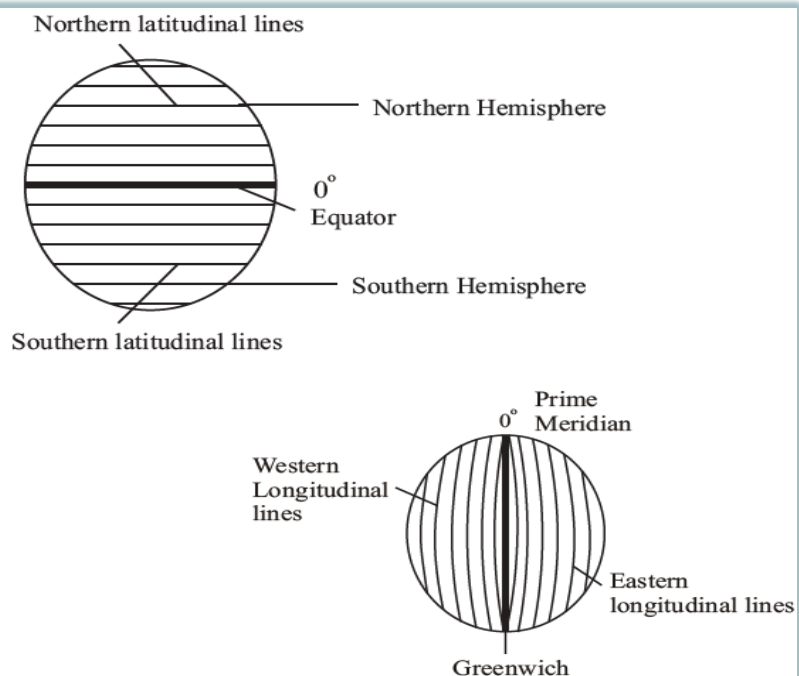
- **Tropic of Cancer (23°30' N)** passes almost through the middle of India.
- **70° East longitude** passes through **Jaisalmer (Rajasthan)**.
- **Sikkim latitude** aligns with **Rajasthan**.
- Time lag **Gujarat–Arunachal Pradesh ≈ 2 hours**.

INDIA AND THE TROPIC OF CANCER & STANDARD TIME



INDIA AND THE TROPIC OF CANCER

- **Latitude: 23°30' N**
- **Divides India: into two halves**
 - **North** → Sub-tropical / Temperate zone
 - **South** → Tropical zone



LATITUDINAL & LONGITUDINAL EXPANSION

- **States it passes through (8):**
 1. Gujarat
 2. Rajasthan
 3. Madhya Pradesh
 4. Chhattisgarh
 5. Jharkhand
 6. West Bengal
 7. Tripura
 8. Mizoram
- **Important towns near Tropic of Cancer:**
 - **Gandhinagar** (closest town)
 - **Jabalpur, Ujjain** also nearby
- **River crossing twice: Mahi River** (origin: MP → flows via Rajasthan, Gujarat → Gulf of Khambhat)
- **Solar position:**
 - On **21 June (Summer Solstice)**, sun is vertically overhead at Tropic of Cancer.
 - Day length **increases northward**.

FACTS

- Tropic passes through **Jharkhand** (earlier part of Bihar).
- **Does not cross:** Uttar Pradesh, Odisha, Manipur, etc.
- **Nearest city to Tropic:** Kolkata (IAS 2003).
- **Delhi** has maximum day length in **June** among major Indian cities.
- **Manipur** lies north of Tropic of Cancer.

STANDARD TIME (IST)

- **Standard Meridian: 82°30' E longitude**
 - Passes through **Mirzapur (UP)** near Prayagraj
- **IST vs GMT: +5 hrs 30 min** ahead of GMT
- **States crossed by 82°30' E meridian:**
 1. Uttar Pradesh
 2. Madhya Pradesh
 3. Chhattisgarh
 4. Odisha
 5. Andhra Pradesh
- **Earth's rotation & Time:**
 - 360° in 24 hrs → **15° = 1 hr, 1° = 4 min**
 - Local time difference between **Gujarat (west) & Arunachal Pradesh (east) ≈ 2 hrs**
- **World Time Zones:** 24 zones, each ≈ 15° longitude wide

FACTS

- **Rewa (MP)** is closest city to IST meridian.
- **Shillong (92°E)** is 38 min ahead of IST.
- **Baghelkhand Plateau (Koriya district, Chhattisgarh)** – intersection of **Tropic of Cancer & 82°30' E (IST)**.
- **Date Line (180° longitude):** +12 hrs eastward, -12 hrs westward → full 24 hr difference.

EXTREME POINTS & BORDERING COUNTRIES

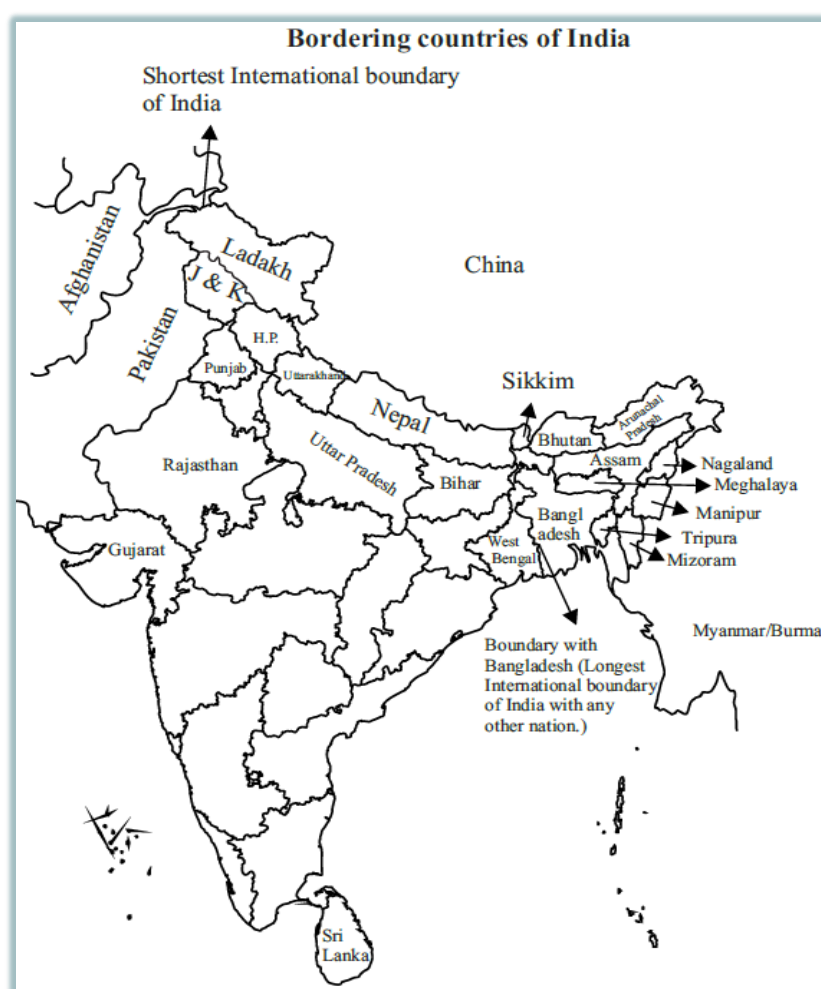
EXTREME POINTS OF INDIA

- **India lies in: Northern & Eastern Hemispheres**
- **Northernmost Point:** Indira Col (Siachen Glacier, Ladakh)
- **Southernmost Point:**

- **Indira Point (Great Nicobar Island, Andaman & Nicobar) → 6°4' N**
- **Mainland southernmost:** Kanyakumari, Tamil Nadu (8°4' N)
- **Easternmost Point:** Kibithu (Arunachal Pradesh)
- **Westernmost Point:** Guhar Moti (Kutch, Gujarat)
- **Special note:**
 - **Kanyakumari** → meeting point of **Bay of Bengal, Arabian Sea, Indian Ocean**
 - **Dong village (Arunachal Pradesh)** → first sunrise in India

FACTS

- **Indira Point** submerged in 2004 Tsunami, but still southernmost.
- **Lucknow (81°E)** is most eastward among major cities (vs Bhopal, Hyderabad, Bengaluru).



BORDERING COUNTRIES OF INDIA

- **Total land boundary: ~15,200 km**
- **Coastline: ~7,516 km**
- **Neighbouring countries (9):**
 1. Pakistan
 2. Afghanistan
 3. China
 4. Nepal
 5. Bhutan
 6. Bangladesh
 7. Myanmar
 8. Sri Lanka (sea boundary via Palk Strait & Gulf of Mannar)
 9. Maldives (sea neighbour, not land)

BORDERS WITH COUNTRIES

- **Pakistan (3,323 km):** Ladakh, J&K, Punjab, Rajasthan, Gujarat

- **Afghanistan (106 km):** Ladakh
- **China (3,488 km):** Ladakh, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh
- **Nepal (1,751 km):** Uttarakhand, UP, Bihar, West Bengal, Sikkim
- **Bhutan (699 km):** Sikkim, West Bengal, Assam, Arunachal Pradesh
- **Bangladesh (4,096.7 km → longest):** West Bengal, Assam, Meghalaya, Tripura, Mizoram
- **Myanmar (1,643 km):** Arunachal Pradesh, Nagaland, Manipur, Mizoram
- **Sri Lanka:** separated by **Palk Strait**
- **Maldives:** southwest of Lakshadweep Islands

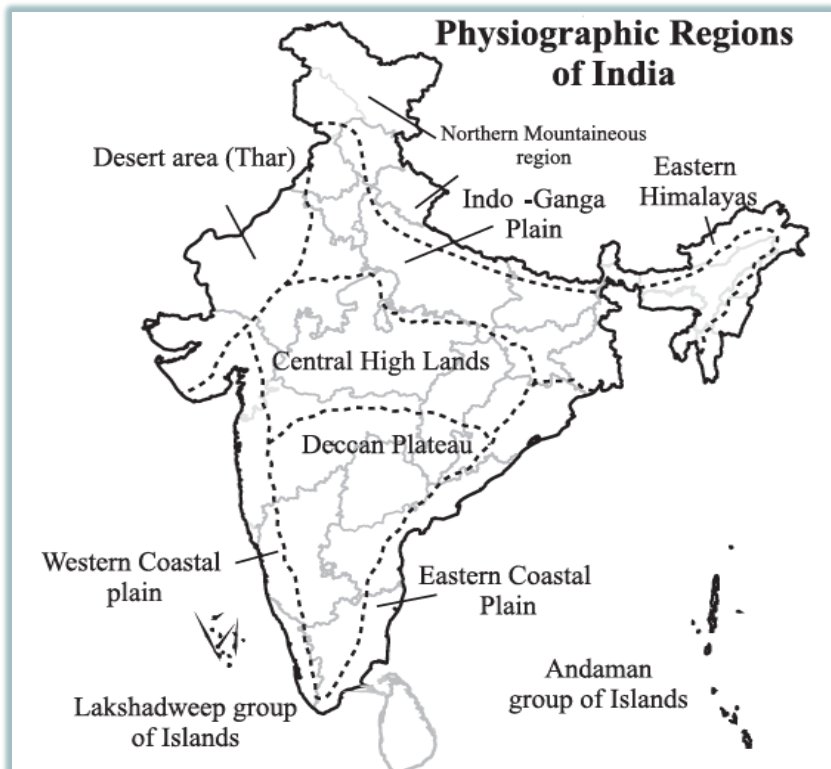
IMPORTANT BORDER LINES

- **Radcliffe Line (1947)** → India–Pakistan
- **McMahon Line (1914)** → India–China
- **Durand Line (1893)** → India–Afghanistan
- **Palk Strait** → India–Sri Lanka
- **Tripura** → bordered by Bangladesh on **three sides**

FACTS

- **Sikkim** shares borders with **Nepal, Bhutan, China**.
- **Bangladesh** → longest land frontier with India.
- **Afghanistan** → shortest land frontier with India.
- **Manipur** shares border only with **Myanmar**, not Bangladesh.
- **Arunachal Pradesh** shares maximum border length with **Myanmar** (among NE states).

PHYSICAL DIVISION OF INDIA NATURAL REGIONS OF INDIA



GEOLOGICAL BACKGROUND

- **India's geological history** linked with **Gondwanaland** (southern supercontinent).
- **Continental drift** → Indian plate broke away, moved northwards → collided with **Eurasian plate**.
- **Tethys Sea** sediments uplifted → **Himalayas** formed.

- Result: **youthful mountains in the north & ancient plateau in the south**.

NATURAL REGIONS OF INDIA

India is divided into **4 major physical divisions**:

1. NORTHERN MOUNTAIN REGION

- Includes **Himalayas & associated ranges**.
- **Youthful topography**: high peaks, deep valleys, fast rivers.
- Important features: **Siachen Glacier, Indira Col, Terai belt (Artesian wells)**.

2. NORTHERN PLAINS

- Formed by **alluvial deposits** of **Ganga, Brahmaputra, Indus systems**.
- **Fertile soils** → **agriculture base**.
- Rich in **groundwater** & dense population.

3. PENINSULAR PLATEAU

- **Oldest landmass of India** (part of Gondwanaland).
- **Hard crystalline rocks** → minerals abundant.
- Plateau regions: **Deccan Plateau, Chotanagpur Plateau, Malwa Plateau, Bundelkhand, Baghelkhand**.

4. COASTAL PLAINS & ISLANDS

- **Western Coastal Plain**: narrow, **Konkan, Kanara, Malabar coasts**.
- **Eastern Coastal Plain**: wide, **Northern Circars, Coromandel Coast**.
- Islands: **Andaman & Nicobar (Bay of Bengal), Lakshadweep (Arabian Sea)**.

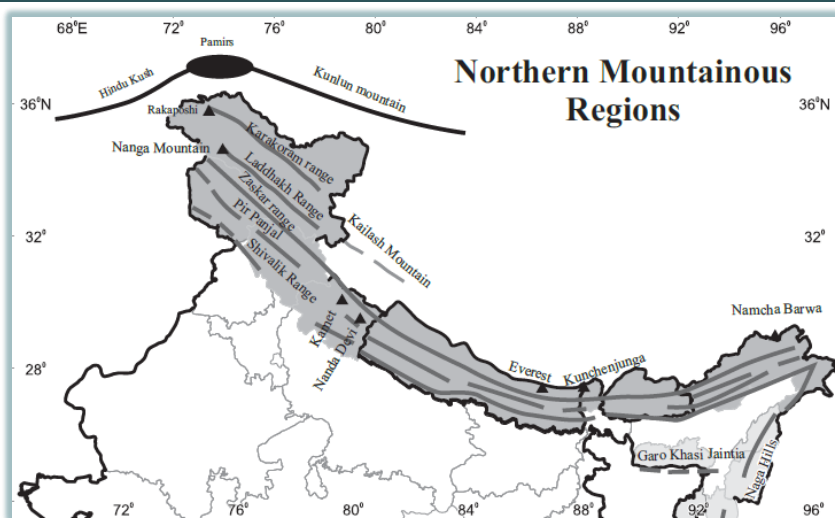
RELIEF DIVISION (DETAILED)

1. **The Himalayas** → Fold mountains, still rising.
2. **Indo-Gangetic-Brahmaputra Plain** → Depositional feature, rich agriculture.
3. **Peninsular Plateau** → Oldest landmass, mineral-rich.
4. **Coastal Plains** → Important for trade, ports, fishing.
5. **Islands** → Strategic + ecological significance.

FACTS

- **Paleomagnetism** studies ancient rock magnetism → supports **continental drift**.
- **Youthful features** (sharp peaks, gorges, glaciers) → Himalayas.
- **Artesian wells** common in **Terai region**.
- **Peninsular Plateau** = part of **Gondwanaland**.
- **Eastern Coastal Plain** wider than **Western**.
- **Andaman & Nicobar** = continuation of **Arakan Yoma (Myanmar hills)**.

NORTHERN MOUNTAINOUS REGION (HIMALAYAS & ASSOCIATED RANGES)



ORIGIN & GENERAL FEATURES

- Formed by collision of Indian Plate with Eurasian Plate (~50 Mya).
- Sediments of Tethys Sea → folded into Himalayas.
- Still rising → frequent earthquakes, landslides.
- Stretch: ~2,400 km long, 250–350 km wide.
- Extent: Indus (W) → Brahmaputra (E).
- Acts as climatic, cultural & political barrier.
- Source of major perennial rivers.

DIVISION OF HIMALAYAS (WEST → EAST)

1. Punjab Himalaya

- Between Indus & Satluj.
- Called Kashmir Himalaya / Himachal Himalaya.
- Famous valleys: Kashmir Valley, Kangra Valley.
- Passes: Banihal, Zoji La, Shipki La.
- Peaks: Nanga Parbat (8,126 m).

2. Kumaon Himalaya

- Between Satluj & Kali rivers.
- States: Uttarakhand.
- Peaks: Nanda Devi, Kamet, Trisul.
- Important glaciers: Gangotri, Yamunotri.
- Passes: Niti, Lipulekh, Mana.

3. Nepal Himalaya

- Between Kali & Tista rivers.
- Contains highest peaks of world:
 - Mt. Everest (8,848.86 m)
 - Kanchenjunga, Makalu, Dhaulagiri, Annapurna.

4. Sikkim Himalaya

- Between Tista & Dihang rivers.
- Peaks: Kanchenjunga (8,586 m, India's highest).
- Passes: Nathu La, Jelep La.

5. Assam / Eastern Himalaya

- Between Dihang & Brahmaputra.
- Features: Dense forests, heavy rainfall, low snowline.
- Hills: Mishmi, Patkai, Naga, Manipur, Mizo hills (Purvanchal).

LONGITUDINAL DIVISIONS OF HIMALAYAS

1. **Trans-Himalaya:** Karakoram, Ladakh, Zaskar ranges.

- Karakoram: Highest peak = K2 (8,611 m).
 - Glaciers: Siachen, Baltoro.
2. **Greater Himalaya (Himadri)**
 - Highest, continuous, snow-clad.
 - Avg. height: 6,000 m.
 - Peaks: Everest, Kanchenjunga, Nanda Devi.
 3. **Lesser Himalaya (Himachal)**
 - Avg. height: 3,500–4,500 m.
 - Valleys: Kashmir, Kangra, Kullu.
 - Hill stations: Shimla, Mussoorie, Nainital.
 4. **Shivalik (Outer Himalaya)**
 - Lowest (900–1,200 m).
 - Composed of unconsolidated sediments.
 - Contains Duns (valleys): Dehra Dun, Patli Dun.

OTHER MOUNTAIN SYSTEMS

- **Purvanchal Hills** (NE extension of Himalayas): Patkai, Naga, Mizo, Manipur hills, Arakan Yoma.
- **Dafla Hills:** Arunachal Pradesh.
- **Garo, Khasi, Jaintia Hills:** Meghalaya plateau.

SNOWLINES & GLACIERS

- **Snowline** (height at which snow is permanent):
 - W. Himalayas: ~4,500 m
 - E. Himalayas: ~3,000 m (due to heavy rainfall).
- Major Glaciers: Siachen (Karakoram), Gangotri, Yamunotri, Zemu (Sikkim).

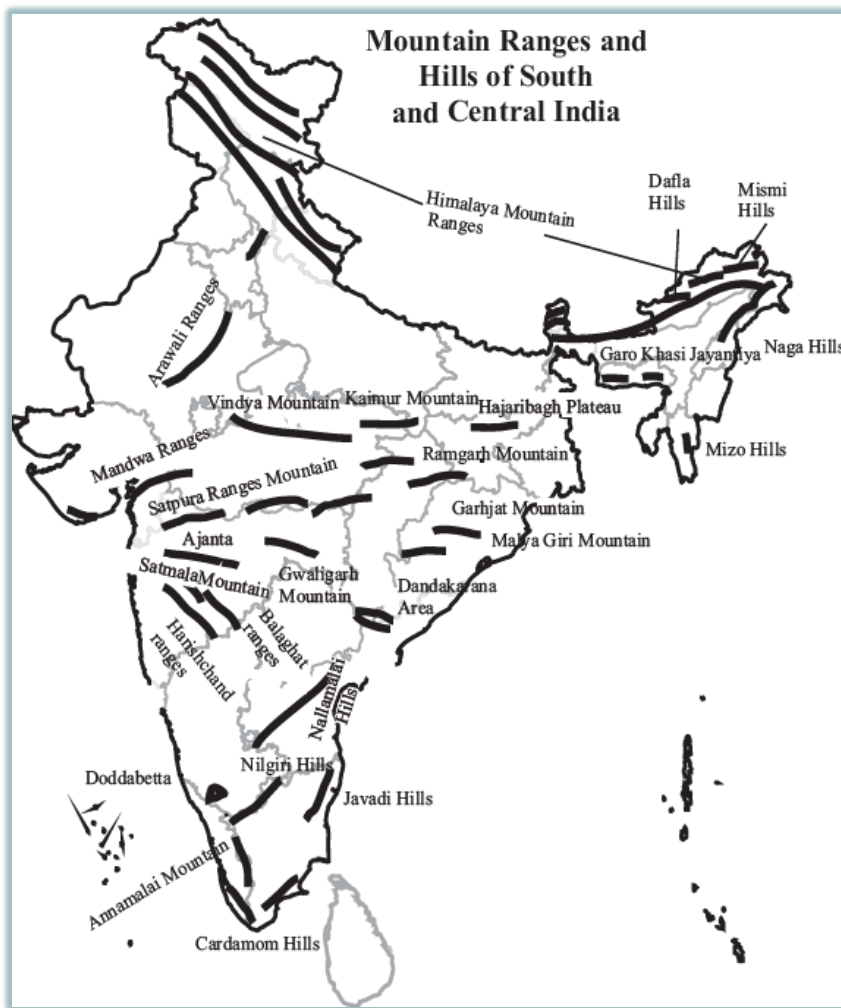
FACTS

- K2 (Godwin Austen) → 2nd highest in world, Karakoram Range.
- Siachen Glacier → World's highest battlefield.
- Nathu La, Jelep La → India–China passes in Sikkim.
- Zoji La → Srinagar–Leh road.
- Khyber & Bolan passes → India–Afghanistan gateways (historic).
- Mawsynram/Cherrapunji (Meghalaya Hills) → highest rainfall.
- Purvanchal Hills = extension of Himalayas into NE India.

MOUNTAIN RANGES AND HILLS OF SOUTH & CENTRAL INDIA

GENERAL FEATURES

- The **Peninsular Plateau** → oldest part of Indian landmass (Archaean rocks, part of Gondwanaland).
- Relief: **Residual mountains, dissected plateaus, rolling hills.**
- Composition: **Igneous & metamorphic rocks** → rich in minerals.
- Character: **Eroded, rounded hills** (vs youthful Himalayas).



MAJOR MOUNTAIN RANGES OF PENINSULAR INDIA

1. ARAVALLI RANGE

- **Oldest fold mountains** of India (Pre-Cambrian).
- Trend: **SW–NE** (Rajasthan → Delhi).
- Highest peak: **Guru Shikhar (1722 m, Mt. Abu)**.
- Significance: Rich in **minerals** (copper, zinc, lead).

2. VINDHYA RANGE

- Trend: **East–West**, mainly in **Madhya Pradesh**.
- Separates **northern plains** from **Deccan plateau**.
- Highest point: **Sad-bhavana Shikhar**.

3. SATPURA RANGE

- Lies south of Vindhyas.
- Includes: **Rajpipla Hills, Mahadeo Hills, Maikal Hills**.
- Highest peak: **Dhupgarh (1350 m, Pachmarhi Hills, MP)**.
- Rich in **forests & minerals**.

4. WESTERN GHATS (SAHYADRI)

- **UNESCO World Heritage site**.
- Stretch: Gujarat–Maharashtra–Goa–Karnataka–Kerala.
- Average height: **1000–1600 m**.
- Peaks:
 - **Anaimudi (2695 m, Kerala)** – highest in South India.
 - **Dodabetta (2637 m, Nilgiris, TN)**.
- Passes: **Bhor Ghat, Thal Ghat, Palghat Gap**.
- Climate role: **Orographic rainfall (SW monsoon)**.

5. EASTERN GHATS

- Discontinuous, eroded ranges along east coast.

- Crossed by rivers: **Mahanadi, Godavari, Krishna, Kaveri**.
- Highest peak: **Jindhagada (1690 m, Andhra Pradesh)**.

6. NILGIRI HILLS

- Junction of **Western & Eastern Ghats**.
- Highest peak: **Dodabetta (2637 m)**.
- Hill station: **Ooty (Udhagamandalam)**.

7. CARDAMOM HILLS

- Southern Kerala.
- Famous for **plantations (cardamom, spices, tea, coffee)**.

8. ANAMALAI HILLS

- Kerala–TN border.
- Peaks: **Anaimudi (2695 m)**.
- Famous for **Periyar Tiger Reserve**.

9. SHEVAROY HILLS

- TN, Salem district.
- Hill station: **Yercaud**.

10. JAVADI HILLS

- Extension of Eastern Ghats, Tamil Nadu.

HILLS OF CENTRAL & EASTERN INDIA

- **Chotanagpur Plateau (Jharkhand, Odisha, Chhattisgarh, WB)**
 - Hills: **Rajmahal, Hazaribagh, Ranchi plateau**.
 - Mineral-rich: **coal, iron, mica, bauxite**.
- **Maikal Hills (Chhattisgarh–MP)**.
- **Ajanta & Satmala Hills (Maharashtra)**.
- **Balaghat Range (MP–Maharashtra border)**.
- **Rajmahal Hills (Jharkhand–Bihar)** → famous for **geological fossils**.

MEGHALAYA PLATEAU HILLS

- Separated from Peninsular plateau by **Garo–Rajmahal Gap**.
- Hills: **Garo, Khasi, Jaintia**.
- Known for **Mawsynram / Cherrapunji** – world's heaviest rainfall.

FACTS

- **Aravallis** → oldest fold mountains in world.
- **Dhupgarh (Satpuras)** → highest peak of Central India.
- **Nilgiris** → meeting point of **Eastern & Western Ghats**.
- **Anaimudi** → highest peak in South India.
- **Palghat Gap** → main link between Kerala & Tamil Nadu.
- **Rajmahal Hills** → known for **Jurassic fossils**.
- **Garo–Khasi–Jaintia Hills** → Meghalaya Plateau.

- **Classical Languages** (recognized by GoI): **Sanskrit, Tamil, Telugu, Kannada, Malayalam, Odia.**

LANGUAGE CENSUS HIGHLIGHTS

- **Hindi** → spoken by ~44% population (2011 census).
- **Bengali** → 2nd most spoken, ~8%.
- **Marathi, Telugu** → ~7% each.
- **English** → widely used in education, business, IT.

FACTS

- **Official language** of India → Hindi (Devanagari).
- **Associate official** → English.
- **Most spoken language** → Hindi.
- **Second most spoken** → Bengali.
- **Classical languages** (6) → Sanskrit, Tamil, Telugu, Kannada, Malayalam, Odia.
- **Santhali** → only tribal language in 8th Schedule.
- **Konkani, Manipuri, Nepali, Bodo, Dogri, Maithili, Santhali** → added later (1992, 2003 amendments).
- **Sanskrit** → least spoken scheduled language.

DRAINAGE SYSTEM OF INDIA

DRAINAGE SYSTEM – OVERVIEW

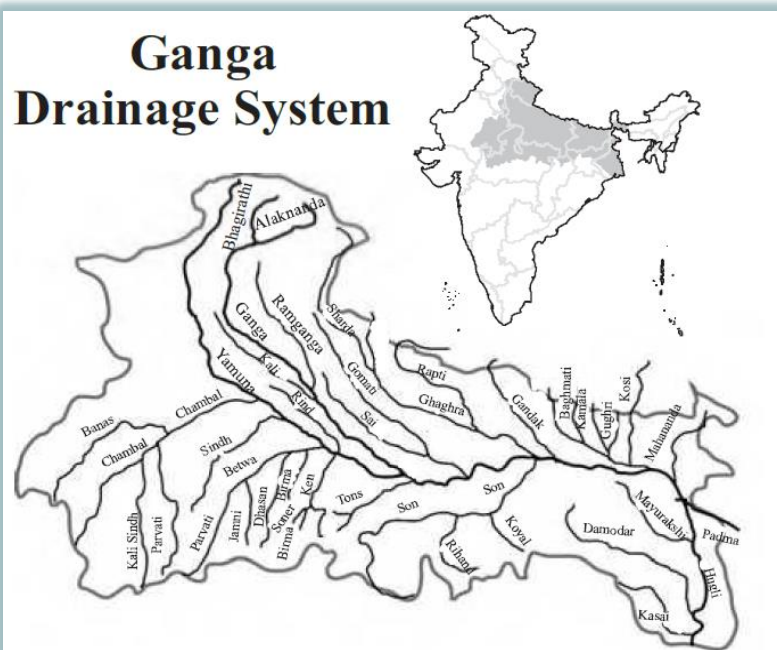
- **Definition:** The pattern formed by rivers & their tributaries.
- Two main types in India:
 1. **Himalayan Rivers** → Perennial, snow-fed + rainfall.
 2. **Peninsular Rivers** → Seasonal, rain-fed, some east-flowing.
- **Drainage patterns:**
 - Dendritic (tree-like) – e.g., Ganga basin.
 - Radial – e.g., rivers from Amarkantak.
 - Trellis – e.g., Chambal, Betwa.
 - Centripetal – e.g., inland drainage in Rajasthan.

HIMALAYAN RIVERS

1. INDUS RIVER SYSTEM

- Origin: **Tibetan Plateau (near Mansarovar, Kailash Range).**
- Length: ~2,880 km (in India ~1,114 km).
- Enters India at **Ladakh** → flows through **J&K** → into Pakistan.
- Tributaries:
 - Left bank: **Jhelum, Chenab, Ravi, Beas, Sutlej** (all 5 rivers of Punjab).
 - Right bank: **Shyok, Gilgit, Zaskar, Hunza.**
- Indus Water Treaty (1960): India can use waters of **Ravi, Beas, Sutlej.**

Ganga Drainage System



2. GANGA RIVER SYSTEM

- Origin: **Gangotri Glacier (Uttarakhand)** → Bhagirathi River.
- Joins **Alaknanda** at Devprayag → becomes **Ganga**.
- Major Tributaries:
 - Left: **Ramganga, Gomti, Ghaghara, Gandak, Kosi.**
 - Right: **Yamuna, Son, Chambal, Betwa, Ken, Tons.**
- Length: ~2,525 km.
- Basin: Covers UP, Bihar, West Bengal, parts of MP, Rajasthan, Jharkhand.
- Forms world's **largest delta (Sundarbans)** before entering Bay of Bengal.

Brahmaputra River System



3. BRAHMAPUTRA RIVER SYSTEM

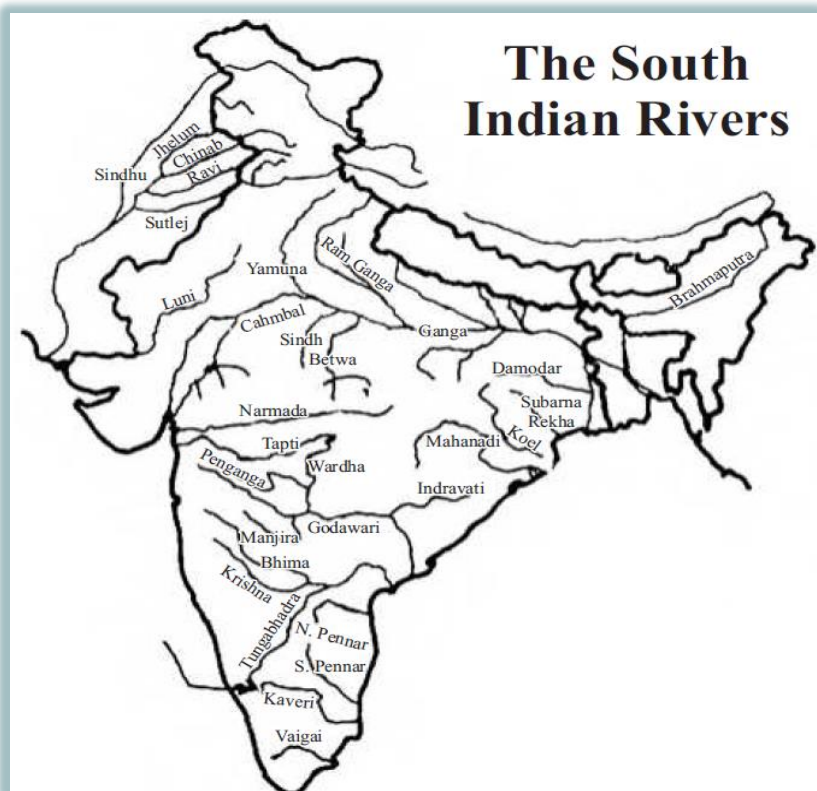
- Origin: **Chemayungdung Glacier (Tibet, near Mansarovar).**
- Known as **Tsangpo** in Tibet.

- Enters India in **Arunachal Pradesh (Siang/Dihang River)**.
- Joins with **Dibang & Lohit** → forms mighty Brahmaputra in Assam.
- Enters Bangladesh → merges with Ganga → forms **Meghna River**.
- Features: Wide, braided channel, prone to floods.
- Length: ~2,900 km (916 km in India).

FACTS

- **Indus** → originates near Mansarovar; longest river of India (including Pakistan).
- **Jhelum** → Wular Lake; Srinagar on its banks.
- **Chenab** → largest tributary of Indus (Punjab = land of 5 rivers).
- **Ganga** → Devprayag confluence (Bhagirathi + Alaknanda).
- **Yamuna** → longest tributary of Ganga (origin: Yamunotri Glacier).
- **Kosi** → “Sorrow of Bihar”.
- **Brahmaputra** → called Tsangpo in Tibet; Dihang in Arunachal.
- **Sundarbans Delta** → largest in world, UNESCO site.

PENINSULAR RIVERS, LAKES & WATERFALLS



PENINSULAR RIVERS – GENERAL FEATURES

- Mostly **seasonal** (rain-fed), except some with dams/reservoirs.
- Flow direction: **Majority eastward into Bay of Bengal**, a few westward into Arabian Sea.
- Patterns: **Radial, Trellis, Dendritic**.

MAJOR EAST-FLOWING RIVERS

- **Mahanadi**
 - Origin: **Chhattisgarh**.
 - Length: ~851 km.

- Drains into **Bay of Bengal** (Paradip port).
- Hirakud Dam.
- **Godavari** (Dakshin Ganga)
 - Origin: **Nasik, Maharashtra (Trimbakeshwar)**.
 - Length: ~1,465 km (longest Peninsular river).
 - Tributaries: **Pravara, Manjra, Indravati, Sabari**.
 - Delta in Andhra Pradesh.
- **Krishna**
 - Origin: **Mahabaleshwar (Maharashtra)**.
 - Tributaries: **Koyna, Bhima, Tungabhadra**.
 - Dams: **Srisailem, Nagarjuna Sagar**.
- **Cauvery (Kaveri)**
 - Origin: **Talakaveri, Brahmagiri Hills (Karnataka)**.
 - Tributaries: **Kabini, Bhavani, Hemavati, Amaravati**.
 - Famous waterfalls: **Shivasamudram Falls**.
 - Known as “**Dakshina Ganga**”.

MAJOR WEST-FLOWING RIVERS

- **Narmada**
 - Origin: **Amarkantak Plateau (MP)**.
 - Flows in **rift valley between Vindhya & Satpura**.
 - Drains into **Arabian Sea (Bharuch Gulf of Khambhat)**.
 - Dams: **Sardar Sarovar, Omkareshwar**.
 - Waterfall: **Dhuandhar Falls (Jabalpur, MP)**.
- **Tapi (Tapti)**
 - Origin: **Satpura Hills (MP)**.
 - Also flows in **rift valley**.
 - Drains into **Arabian Sea** (near Surat).
- **Mandovi & Zuari** (Goa).
- **Sharavathi** → Jog Falls (Karnataka).
- **Periyar** (Kerala).

INLAND DRAINAGE

- **Rajasthan (Thar Desert)** → Luni River → disappears in Rann of Kutch.
- Some salt lakes → **Sambhar Lake (Rajasthan)**.

IMPORTANT LAKES

- **Freshwater Lakes**
 - **Wular Lake (J&K)** → largest freshwater lake in India.
 - **Loktak Lake (Manipur)** → floating phumdis, Keibul Lamjao NP.
 - **Bhimtal, Nainital (Uttarakhand)**.
- **Saltwater Lakes**
 - **Chilika Lake (Odisha)** → largest brackish water lagoon.
 - **Pulicat Lake (TN-AP)**.
 - **Sambhar Lake (Rajasthan)** → salt production.

IMPORTANT WATERFALLS

- **Jog Falls** (Sharavathi, Karnataka) – highest plunge waterfall.

WORLD GEOGRAPHY

UNIVERSE

STRUCTURE OF THE UNIVERSE

- **Universe** = All matter, energy, planets, stars, galaxies, intergalactic space.
- **Age**: ~13.7–13.8 billion years (based on expansion rate).
- **Origin**: Explained by **Big Bang Theory**.

KEY ASTRONOMICAL UNITS

- **Light Year**: Distance travelled by light in 1 year in vacuum = 9.461×10^{12} km.
- **Astronomical Unit (AU)**: Avg. Earth–Sun distance = **149.6 million km**.
- **Parsec**: 3.26 light years (used for galactic distances).

THEORIES

- **Big Bang Theory**
 - Proposed by **George Lemaitre (1927)**, later supported by **Robert Wagner (1967)**.
 - Universe originated from hot dense singularity → sudden explosion → expansion.
 - Explains formation of galaxies, solar system.
- **Chandrasekhar Limit**
 - Mass limit of a white dwarf = **1.44 solar masses**.
 - Beyond this → collapses into **Neutron Star** or **Pulsar**.

GALAXIES & STARS

- **Galaxy** = massive system of stars, gas, dust bound by gravity.
- **Milky Way Galaxy**: classified as **Spiral Galaxy**.
- **Nearest Galaxy**: Andromeda (M31).
- **Nearest Star**: Proxima Centauri (4.24 light years).
- **Nebula**: cloud of gas & dust, star-forming regions (e.g., **Orion Nebula**).

CONSTELLATIONS

- Imaginary star patterns in sky.
- Recognized by International Astronomical Union: **88 constellations** (1922).
- **Great Bear (Ursa Major, Saptarishi)** → used to locate **Pole Star**.
- **Orion (Hunter)** → visible in winter, contains Orion Nebula.

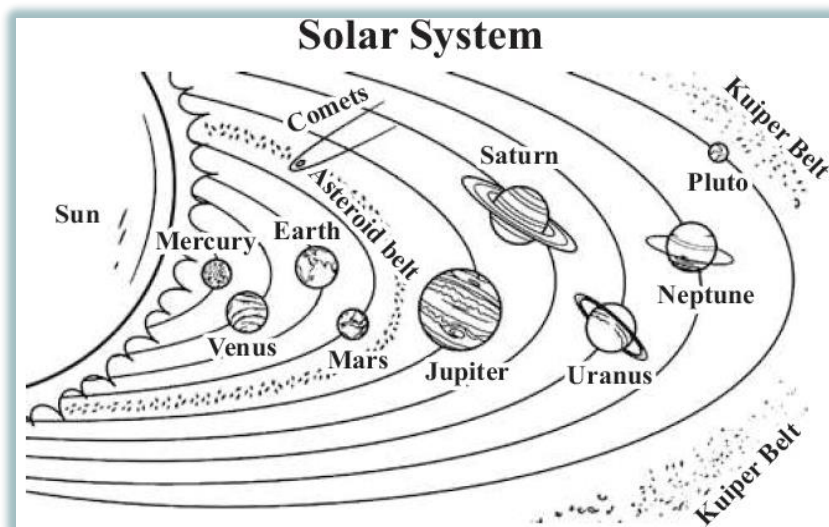
OBSERVATIONS

- Stars appear to revolve around **celestial poles**, parallel to celestial equator.
- **Celestial Equator** → projection of Earth's equator onto sky.
- In space (no atmosphere) → sky appears **black**, not blue.

FACTS

- **Big Bang Theory** → origin of Universe.
- **Milky Way** → Spiral Galaxy.
- **Nearest Star** → Proxima Centauri.
- **Constellation for Pole Star** → Great Bear (Saptarishi).
- **Chandrasekhar Limit** → 1.44 solar masses.
- **Orion Nebula** → visible, star-forming region in Milky Way.

THE SOLAR SYSTEM



FORMATION & STRUCTURE

- **Age**: ~4.6 billion years.
- Components: **Sun (star)** at center, **8 planets**, dwarf planets, moons, asteroids, comets, meteoroids, cosmic dust.
- **Nebulae** → not part of Solar System (they are interstellar gas/dust).

PLANETS (IN ORDER FROM SUN)

1. Mercury
 2. Venus
 3. Earth
 4. Mars
 5. Jupiter
 6. Saturn
 7. Uranus
 8. Neptune
- Earlier **Pluto** was included, but in **IAU 2006**, it was reclassified as a **dwarf planet**.

PLANETARY SIZES (DIAMETER)

- **Jupiter** – 142,984 km (largest).
- **Saturn** – 120,536 km.
- **Uranus** – 51,118 km.
- **Neptune** – 49,528 km.
- **Earth** – 12,756 km (densest planet).
- **Venus** – 12,104 km.
- **Mars** – 6,792 km.
- **Mercury** – 4,879 km (smallest).

PERIOD OF REVOLUTION (AROUND SUN)

- Mercury – 88 days.
- Venus – 225 days.

- Earth – 365 days.
- Mars – 687 days.
- Jupiter – 11.86 years.
- Saturn – 29.46 years.
- Uranus – 83.7 years.
- Neptune – 164.9 years (longest among 8 planets).
- Pluto (dwarf planet) – 247.9 years.

SATELLITES OF PLANETS (AS OF FEB 2024)

- Mercury – 0
- Venus – 0
- Earth – 1 (Moon)
- Mars – 2 (Phobos, Deimos)
- Jupiter – 95 (Galilean moons: **Io, Europa, Ganymede, Callisto**)
- Saturn – 146 (largest = **Titan**, 5150 km)
- Uranus – 27
- Neptune – 14

KEY CONCEPTS

- **Heliocentric Theory:** Sun is center, Earth revolves around it.
 - Propounded by **Nicolaus Copernicus (1514, "Commentariolus")**.
 - **Varahamihira (6th century India)** had already proposed similar idea.
- **Earth = densest planet** in Solar System.
- **Venus** → hottest planet (464°C) due to **greenhouse effect**.
- **Ganymede (moon of Jupiter)** → largest natural satellite in Solar System.
- **Goldilocks Zone** → habitable zone where liquid water exists; Earth is the only planet in it.

FACTS

- Solar System age → **4.6 billion years**.
- **Copernicus** → Heliocentric theory (supported by Varahamihira earlier).
- **Earth = densest planet**.
- **Smallest planet** → Mercury (4,879 km).
- **Largest planet** → Jupiter (142,984 km).
- **Coldest planet** → Neptune (–200°C).
- **Longest revolution** → Neptune (165 yrs), but Pluto (dwarf) = 248 yrs.
- **Ganymede (Jupiter's moon)** → largest in Solar System.
- **Galileo (1610)** → discovered Jupiter's 4 major moons.

THE SUN

BASIC FACTS

- **The Sun** → a **medium-sized star (yellow dwarf)** in the Milky Way.
- Distance from Earth → **~149.6 million km (1 AU)**.
- Diameter → **~1.39 million km (109× Earth)**.
- Mass → **~333,000× Earth**.
- Composition → **~75% hydrogen, ~24% helium, ~1% other elements**.
- Provides **99.8% of Solar System's total mass**.

STRUCTURE OF THE SUN

1. **Core**
 - Central region, site of **nuclear fusion**.
 - Hydrogen → Helium fusion releases energy.
 - Temperature: **~15 million °C**.
2. **Radiative Zone**
 - Energy transfer by radiation.
3. **Convective Zone**
 - Energy carried upward by convection currents.
4. **Photosphere**
 - Visible surface of Sun.
 - Temperature: **~5,500 °C**.
5. **Chromosphere**
 - Above photosphere, seen during solar eclipse.
6. **Corona**
 - Outer layer, visible as white halo during eclipse.
 - Very hot (**~1–2 million °C**).

SOLAR PHENOMENA

- **Sunspots** → Dark, cooler regions on photosphere due to magnetic activity.
- **Solar Flares** → Sudden explosions releasing charged particles.
- **Solar Wind** → Stream of charged particles (protons, electrons) → affects Earth's magnetosphere (Auroras).
- **Solar Cycle** → **~11 years** (periodic increase/decrease of sunspots).

IMPORTANCE

- Source of **heat & light** → drives Earth's climate.
- Fusion releases enormous energy (Einstein's equation **$E=mc^2$**).
- Solar radiation powers hydrological cycle, winds, photosynthesis.

FACTS

- Sun's energy source → **Nuclear fusion (Hydrogen → Helium)**.
- Photosphere → **visible surface** of Sun.
- Corona → seen during **total solar eclipse**.
- Solar cycle length → **11 years**.
- Sunspots → **cooler & dark regions**.
- Sun's diameter → **109× Earth's diameter**.
- Fraction of Solar System's mass in Sun → **99.8%**.

THE MERCURY

BASIC FACTS

- **Mercury** → **smallest planet** in the Solar System and **nearest to the Sun**.
- Diameter → **4,879 km**.
- Distance from Sun → **57.9 million km (0.39 AU)**.
- Revolution around Sun → **88 days** (shortest of all planets).
- Rotation (day length) → **~59 Earth days**.
- Satellites → **None** (like Venus).

SURFACE & ATMOSPHERE

- **No significant atmosphere** → cannot retain heat.
- Extreme temperature variation:
 - Daytime: up to **430°C**.
 - Night: as low as **-180°C**.
 - Avg. ~167°C.
- **Albedo** (reflectivity): very low **0.1** vs Earth's **0.3** → appears less bright despite strong sunlight.

EXPLORATION

- **Mariner 10 (NASA, 1974–75)** → first spacecraft to fly by Mercury; photographed surface.
- Revealed → **mountains, plains, craters** (similar to Moon).

ORBITAL FEATURES

- Highly **elliptical orbit** → eccentricity ~0.206.
- Moves fastest in orbit due to Sun's strong gravity → highest kinetic energy at perihelion.

FACTS

- **Smallest planet** → Mercury (not Pluto, since 2006 IAU decision).
- **Revolution** → 88 days (shortest).
- **No moons** → Mercury & Venus only.
- **High temperature fluctuations** due to lack of atmosphere.
- **Albedo** → 0.1 (low brightness).
- **NASA's Mariner 10** → photographed Mercury.

THE VENUS

BASIC FACTS

- **Venus** → 2nd planet from Sun, ~**108.2 million km (0.72 AU)** away.
- Revolution period → **224.7 days**.
- Rotation → **retrograde (east to west)** like Uranus.
- Diameter → **12,104 km** (close to Earth's size).
- **No natural satellites** (like Mercury).

TEMPERATURE & ATMOSPHERE

- **Hottest planet** of the Solar System → Avg. **464°C** (737 K).
- Atmosphere → **96.5% CO₂** (strong greenhouse effect).
- Pressure = 90× Earth's → "**Pressure cooker effect**".
- Makes **life impossible** due to high CO₂ + heat.

VISIBILITY & NICKNAMES

- **Brightest planet** in night sky.
- Called **Morning Star & Evening Star**.
- Also called "**Twin Sister of Earth**" (similar size, mass, composition).

EXPLORATION

- **Magellan Spacecraft (NASA, 1989)** → studied Venus (1990–1994).
- Revealed surface with **mountains, plains, volcanoes**.

FACTS

- **Hottest planet** → Venus (not Mercury).
- Atmosphere → **96.5% CO₂**.
- Called **Morning Star & Evening Star**.
- Known as **Earth's twin**.
- Rotation → **retrograde** (east to west).
- **No moon** (like Mercury).
- Explored by **Magellan spacecraft**.

THE EARTH

BASIC FACTS

- **3rd planet from Sun**.
- Distance from Sun → **149.6 million km (1 AU)**.
- Diameter → **12,756 km** (equatorial); Polar diameter slightly less (oblate spheroid).
- Circumference → **~40,075 km** (equatorial).
- Rotation → **23 hr 56 min 4 sec** (sidereal day).
- Revolution → **365 days 5 hr 48 min 45 sec** (tropical year).

SHAPE & MOTION

- Shape → **Geoid (oblate spheroid)**: bulging at equator, flattened at poles.
- **Axial Tilt: 23.5°** → causes **seasons**.
- **Equinox** → Sun overhead at Equator on **21 March & 23 September** → day & night equal.
- **Summer Solstice (21 June)** → Sun overhead at **Tropic of Cancer** (23.5°N).
- **Winter Solstice (22 Dec)** → Sun overhead at **Tropic of Capricorn** (23.5°S).
- **Aphelion (July 4)** → farthest from Sun.
- **Perihelion (Jan 3)** → closest to Sun.

LIFE-SUPPORTING FEATURES

- Only known **habitable planet** (within **Goldilocks Zone**).
- Atmosphere: **78% Nitrogen, 21% Oxygen, 1% Argon & others (incl. CO₂ ~0.03%)**.
- **Moderate temperature** due to atmosphere & water cycle.
- Presence of **liquid water** → essential for life.

INTERNAL STRUCTURE

1. **Crust** – outermost, solid (continental & oceanic).
 - Continental crust → **granitic**; thicker (30–70 km).
 - Oceanic crust → **basaltic**; thinner (~5–10 km).
2. **Mantle** – ~2,900 km thick, made of **silicate rocks rich in Fe & Mg**.
 - **Asthenosphere**: partially molten, causes plate movement.
3. **Core** – outer (liquid iron & nickel) + inner (solid iron & nickel).
 - Generates **Earth's magnetic field**.
 - Temp: **~5,000–6,000 °C**.

GRAVITATIONAL & ORBITAL CHARACTERISTICS

- **Acceleration due to gravity:** 9.8 m/s^2 .
- Escape velocity → **11.2 km/s**.
- Earth's surface area → 510 million sq. km (29% land, 71% water).
- Earth's density → **5.52 g/cm^3** (highest among planets).

MOTIONS OF EARTH

1. **Rotation** (on axis)
 - Causes **day & night**.
 - Responsible for **apparent movement of Sun & stars**.
2. **Revolution** (around Sun)
 - With axial tilt, causes **seasons**.
 - Results in **variation in day length & solar intensity** across latitudes.

FACTS

- **Shape of Earth** → Geoid (oblate spheroid).
- **Day-night** → caused by Earth's **rotation** on axis.
- **Seasons** → due to **axial tilt (23.5°) + revolution around Sun**.
- Sun overhead at **Equator** → 21 March & 23 Sept (Equinox).
- **Aphelion (4 July)** = farthest, **Perihelion (3 Jan)** = nearest.
- Earth's density = **5.52 g/cm^3** (densest planet).

THE MARS

BASIC FACTS

- **4th planet from Sun** (after Earth).
- Distance from Sun → **228 million km (1.52 AU)**.
- Diameter → **6,792 km**.
- Rotation → **24.7 hrs** (close to Earth's day).
- Revolution → **686.98 days (~1.88 years)**.
- Axial tilt → **$25^\circ 12'$** (very similar to Earth's 23.5°).

SURFACE & ATMOSPHERE

- **Atmosphere composition:** CO_2 95.1%, N_2 2.59%, Argon 1.94%, O_2 0.16%, traces of water.
- Avg. temperature → **-65°C (208 K)**.
- Appearance → **Red Planet** (due to iron oxide on surface).
- Has **polar ice caps** (frozen water).

EXPLORATION

- **Mars Reconnaissance Orbiter (NASA, 2006)** → detected hydrated minerals & liquid water.
- **Phoenix Mars Lander (NASA, 2008)** → found **ice on Mars**.
- **Curiosity Rover (NASA, 2012)** → studying possibility of life, water, climate, geology.
- **Mangalyaan (MoM, ISRO, 2013)** → India's **first Mars mission**, inserted into orbit on 24 Sept 2014; India joined **elite Mars club** (USA, Russia, EU).

LANDFORMS

- **Olympus Mons (Nix Olympia)** → largest volcano in Solar System; **$\sim 3\times$ height of Mt. Everest**.

Moons

- Two small moons: **Phobos & Deimos**.

FACTS

- Mars → **Red Planet** (iron oxide).
- **Day length & tilt** → close to Earth (24.7 hrs, tilt $25^\circ 12'$).
- Avg. temp = -65°C .
- Has **polar ice caps & water traces** (key to habitability).
- **Olympus Mons** = tallest volcano in Solar System.
- Moons → Phobos & Deimos.
- **Mangalyaan (2013, ISRO)** → put India in Mars club.
- NASA's **MRO, Phoenix Lander, Curiosity Rover** → major missions.

THE JUPITER

BASIC FACTS

- **Largest planet** in the Solar System.
- Diameter → **142,984 km**.
- Mass → **$318\times$ Earth**.
- Revolution period → **11.86 years** around the Sun.
- Rotation → fastest among planets (just under **10 hours**).
- Composition → Mainly **Hydrogen & Helium**, with traces of methane, ammonia, ethane, hydrogen deuteride.

MOONS

- Total moons (Feb 2024) → **95 known satellites**.
- **Galilean Moons** (discovered by Galileo in 1610): **Io, Europa, Ganymede, Callisto**.
 - **Ganymede** → largest moon in Solar System.

RINGS

- Jupiter has a **faint ring system** discovered by **Voyager 1 (1979)**.
- Rings are **dust & silicate particles**, almost transparent.
- Known as **Jovian Rings**.

EXPLORATION

- **Voyager 1 (1977)** → explored Jupiter system, detailed Europa's icy surface.
- Speculation: **Europa may have liquid oceans** (60–150 km deep) → possible habitability.

DISTINCT FEATURES

- Known as **"Miniature Solar System"** due to size and many moons.
- Fastest spinning planet in Solar System.
- Produces strong **magnetic field** and **Great Red Spot** (giant storm).

FACTS

- **Largest planet** → Jupiter (diameter 142,984 km).
- **Heaviest planet** → Jupiter .
- **Galilean moons** discovered by Galileo (1610).
- **Ganymede** → largest moon in Solar System.
- Jupiter's rings discovered by **Voyager 1 (1979)**, made of **silicates & dust**.
- Revolution = 11.86 years; Rotation = fastest (~10 hrs).

- Cassini → launched 1997, reached Saturn 2004.
- Saturn appears **yellow** in night sky.
- Magnetic field → 578× Earth's.

THE SATURN

BASIC FACTS

- **6th planet from the Sun**, 2nd largest in Solar System .
- Diameter → **120,536 km** .
- Revolution → **29.4 years** around Sun .
- Rotation → **10 hrs 34 min** .
- Axial tilt → **26.7°** .
- Avg. surface temperature → **-140°C** .

ATMOSPHERE & MAGNETIC FIELD

- Atmosphere: Mainly **Hydrogen & Helium**, with methane, ethane, ammonia, hydrogen deuteride .
- Magnetic field → **578 times Earth's** .

RINGS

- Saturn = **most famous for rings** → system of **7 rings** .
- Rings **A, B, D, E** visible from Earth .
- Other gas giants (Jupiter, Uranus, Neptune) also have rings, but Saturn's are most prominent .
- Appears **yellowish** when seen from Earth .

MOONS

- Total moons → **146** (highest among planets) .
- **Titan** → largest moon (diameter **5,150 km**) .
- **Rhea** → 2nd largest (diameter **1,528 km**) .
- Other satellites: **Dione, Tethys, Mimas, Enceladus, Iapetus, Helene, Atlas, Phoebe** .
- **Phoebe** → orbits Saturn opposite to rotation (retrograde) .

EXPLORATION

- **Cassini probe** launched **15 Oct 1997**, reached Saturn in **June 2004** .
- Provided detailed study of Saturn's rings, moons, and atmosphere.

DISTINCT FEATURES

- Known as "**Globe of Gases**" & "**Galaxy-like Planet**" .
- Warmer than Neptune, though farther than Jupiter .

FACTS

- Saturn's revolution → **29.4 years**.
- **Only planet visibly famous for rings** → Saturn.
- Rings discovered: **7 (A, B, D, E visible)**.
- Largest moon = **Titan**; 2nd largest = **Rhea**.
- Total moons = **146** (highest in Solar System).

URANUS, NEPTUNE & PLUTO

URANUS

- **7th planet** from the Sun, **3rd largest** in Solar System .
- **Axial tilt** → **97.77°** (rotates on its side) .
- Rotation → **17 hrs 14 min** (retrograde; rotates east to west) .
- Revolution → **84 years** around Sun .
- Sun rises in **West** and sets in **East** .
- Atmosphere: **Hydrogen, Helium, Methane, Ammonia** .
- Appears **blue-green** (due to methane absorption of red light) .
- **27 moons**, has **planetary rings** (like Saturn) .

NEPTUNE

- **8th planet** (farthest true planet after Pluto's reclassification in 2006).
- **Discovered by Johann Gottfried Galle** on **23–24 Sept 1846** .
- Rotation → **16 hrs 6 min** .
- Revolution → **164.9 years** .
- **Axial tilt** → **28.3°** .
- **Coldest planet** in Solar System: **~-200°C** .
- Atmosphere: **Hydrogen, Helium, Methane** .
- Moons → **14 known**; main: **Triton, Nereid** .
- Became **farthest planet after 2006**, when **Pluto was reclassified as a dwarf planet** .

PLUTO (DWARF PLANET)

- **Discovered by Clyde Tombaugh** in **1930** .
- Earlier: 9th planet, **smallest** (diameter **2376 km**) .
- **Reclassified as Dwarf Planet in 2006** (IAU, Prague, 14–25 Aug 2006) due to orbit overlapping Neptune's .
- Revolution → **247.9 years** .
- Moons → **5 known**: **Charon (largest)**, Styx, Nix, Kerberos, Hydra .
- Orbit → **highly elliptical**, sometimes brings Pluto closer than Neptune .

FACTS

- **Uranus** → 7th planet, axial tilt **97.77°**, **retrograde rotation**, **27 moons**, blue-green color.
- **Neptune** → discovered in **1846** by **Johann Galle**, **coldest planet (-200°C)**, 14 moons (Triton & Nereid), revolution **164.9 years** .
- **Pluto** → discovered by **Clyde Tombaugh (1930)**, **reclassified as dwarf planet in 2006** (IAU, Prague), revolution **247.9 years**, 5 moons (Charon largest, others: Styx, Nix, Kerberos, Hydra) .
- **Farthest planet**:
 - Earlier → **Pluto**.
 - Since 2006 → **Neptune** (Pluto demoted) .

THE MOON

BASIC FACTS

- **Earth's only permanent natural satellite .**
- Diameter → **3,475 km.**
- Distance from Earth: **Min 363,300 km (Perigee); Max 405,500 km (Apogee) .**
- **Orbital period = Rotational period → 27 days, 7 hrs, 43 min** (synchronous rotation) .
- Because of this, we always see **the same face of the Moon.**
- We can see **~59% of the lunar surface** due to tilt and libration .

PHYSICAL & ENVIRONMENTAL CONDITIONS

- No **atmosphere** → no sound transmission on Moon .
- Surface temperature extremes due to lack of atmosphere.
- Appearance changes with orbit → phases of Moon.

ECLIPSES & TIDES

- **Solar Eclipse** → occurs at **New Moon**, when Moon is between Sun & Earth .
- **Lunar Eclipse** → occurs at **Full Moon**, when Earth comes between Sun & Moon .
- **Perigee** (closest to Earth): stronger gravitational pull → **High tides.**
- **Apogee** (farthest from Earth): weaker pull → **Low tides** (20% smaller than high tides) .
- Example: **Lunar Perigee on 22 Dec 1999** → Moon appeared brightest .

HUMAN EXPLORATION

- **First manned mission: Apollo-11 (USA, NASA).**
 - Launched: **16 July 1969** from Cape Kennedy, Florida.
 - Landed: **20 July 1969** on **Sea of Tranquility.**
 - Astronauts: **Neil Armstrong, Edwin (Buzz) Aldrin, Michael Collins.**
 - Armstrong & Aldrin walked on Moon; Collins remained in orbit .
 - Returned: **24 July 1969**, Pacific Ocean .

KEY TERMS

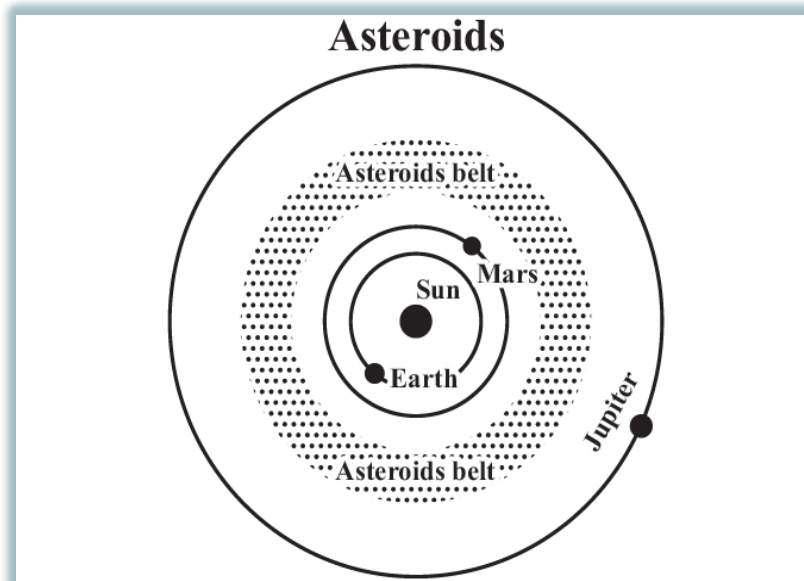
- **Perigee** → closest approach (~363,300 km).
- **Apogee** → farthest distance (~405,500 km).
- **Lunar Mare (e.g., Sea of Tranquility)** → dark basalt plains visible from Earth .

FACTS

- Diameter = **3,475 km.**
- Distance → 363,300 km (Perigee) – 405,500 km (Apogee).
- Orbital/rotational period = **27d 7h 43m.**
- Always shows **same face to Earth.**
- Solar eclipse → New Moon; Lunar eclipse → Full Moon.
- Perigee = High tides; Apogee = Low tides.
- Apollo-11 landed on **Sea of Tranquility** (20 July 1969).

- First man on Moon → **Neil Armstrong**, followed by **Buzz Aldrin.**

ASTEROIDS



DEFINITION

- **Asteroids = small rocky & metallic celestial bodies** revolving around the Sun .
- Also called **planetoids** or **minor planets.**

LOCATION

- Found mainly **between Mars & Jupiter** → region called **Asteroid Belt** .
- This belt separates **inner (terrestrial)** and **outer (giant)** planets.

SIZE

- Mostly small in size, but some can reach up to **1000 km diameter.**
- Example: **Ceres** (largest asteroid, also classified as a **dwarf planet**).

CLASSIFICATION

1. **Main Belt Asteroids**
 - Located between **Mars & Jupiter.**
 - Prominent asteroids: **Hungaria, Flora, Phocaea, Eos, Themis, Cybele, Hilda, Koronis .**
2. **Near Earth Asteroids (NEAs)**
 - Cross Earth's orbit.
 - Important groups: **Aten, Apollo, Amor, Atira .**
3. **Trojans**
 - Located near **Lagrange points of Jupiter's orbit .**

DISTINCTION FROM COMETS

- **Asteroids:** made of **rock & metals**, solid, no tail.
- **Comets:** made of **ice, dust, gases**; develop **glowing tail** when near the Sun (due to sublimation) .
- Asteroids → closer to Sun; Comets → originate from **Kuiper Belt & Oort Cloud** .

FACTS

- **Asteroid Belt** lies between **Mars & Jupiter** (NOT Jupiter & Saturn).
- Largest asteroid = **Ceres (~1000 km)**.
- Examples: Hungaria, Flora, Phocaea, Eos, Themis, Cybele, Hilda, Koronis (Main Belt).
- NEAs → Aten, Apollo, Amor, Atira.
- Trojans → near **Jupiter's Lagrange points**.
- Difference from comets → Asteroids are rocky/metallic; Comets are icy & form **tails near Sun**.

COMETS AND METEORS

COMETS

- **Definition:** Small icy bodies that, when near the Sun, heat up and release gases → form a glowing **coma (atmosphere)** and a **tail**.
- **Orbit:** Highly **elliptical**; revolve around Sun.
- **Tail:** Always points **away from Sun** due to **solar wind & radiation pressure**.
 - Two types of tails:
 1. **Dust tail**
 2. **Ionized (gas) tail**

FAMOUS COMETS

- **Halley's Comet**
 - **Period:** ~76 years.
 - Short-period comet.
 - Last appeared in **1986**, will return in **2061**.
- **Hale-Bopp Comet**
 - Discovered on **23 July 1995** by **Alan Hale (New Mexico) & Thomas Bopp (Arizona)**.
 - Period: **2534 years** (long-period comet).
 - ~**1000 times brighter than Halley's comet**.
- **Shoemaker-Levy 9**
 - Discovered **24 March 1993** by **E.M. Shoemaker, Carolyn Shoemaker, David Levy**.
 - Collided with **Jupiter (16–22 July 1994)** → first direct observation of extraterrestrial collision.
- **67P/Churyumov-Gerasimenko**
 - Destination of **Rosetta Mission (ESA)**.
 - Rosetta launched **2 March 2004**, spacecraft landed on comet **6 Aug 2014**.

METEORS

- **Definition:** A **piece of matter from outer space** that enters Earth's atmosphere.
- Appears as **bright streak of light** (shooting star).
- Glows due to **air friction & heat** on entry.
- **Most burn completely** before reaching Earth's surface.

RELATED TERMS

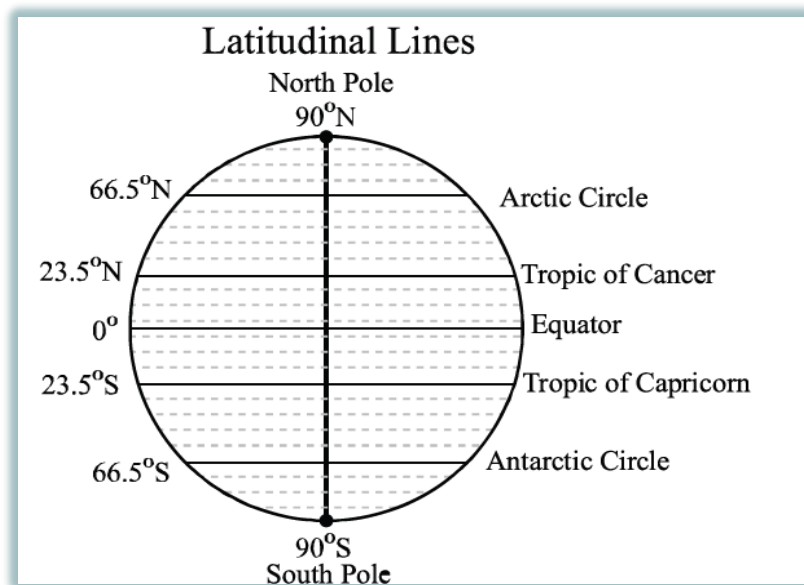
- **Meteoroid** → a small fragment of rock in space.
- **Meteor** → when meteoroid enters Earth's atmosphere and glows.
- **Meteorite** → when fragment survives and reaches Earth's surface.

FACTS

- Comet's tail → **always away from Sun** due to solar radiation.
- **Halley's Comet** → ~76 years periodicity.
- **Hale-Bopp** → discovered 1995, 2534 years period, 1000× brighter than Halley.
- **Shoemaker-Levy 9** → collided with Jupiter (1994).
- **Rosetta Mission** → first landing on a comet (67P, 2014).
- Meteor = shooting star, burns in atmosphere; Meteorite = reaches Earth.

EARTH

LATITUDE



DEFINITION

- **Latitude:** The angular distance of a place north or south of the **Equator**, measured in degrees.
- Represented as **parallels** (imaginary horizontal lines) that circle the globe.
- Measured from **0° at Equator** to **90° at poles** (North & South).

CHARACTERISTICS

- **Equator** → 0° latitude, longest parallel.
- **Poles** → 90° North & 90° South, points not circles.
- **Total number of latitudes** = 181 (from 90°N to 90°S, including equator).
- Each degree of latitude ≈ **111 km** apart.

CLIMATIC ZONES (BASED ON LATITUDE)

1. **Torrid Zone** (0°–23.5° N & S)
 - Receives **direct sun rays** → hottest zone.
2. **Temperate Zones** (23.5°–66.5° N & S)
 - Moderate climate.
3. **Frigid Zones** (66.5°–90° N & S)
 - Coldest regions, near poles.

MAJOR LATITUDES

- **Equator** → 0°
- **Tropic of Cancer** → 23.5° N
- **Tropic of Capricorn** → 23.5° S

5. South America – 554 m
6. Australia – 330 m
7. Europe – 302 m

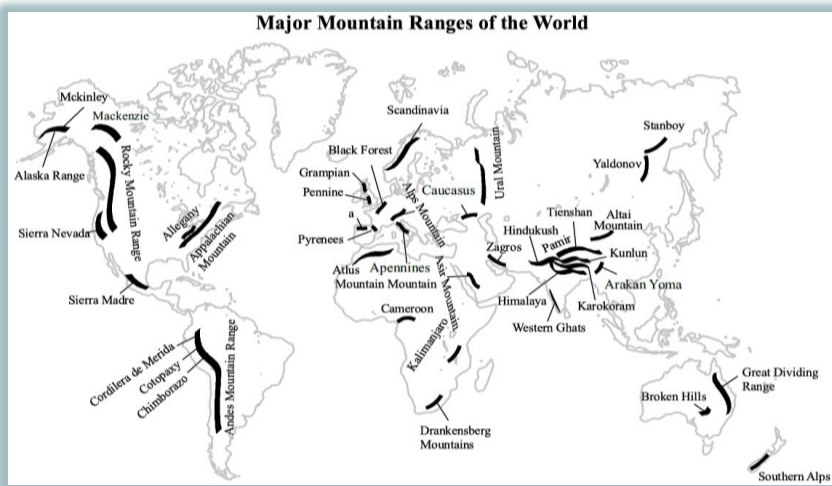
FACTS

- **Largest continent:** Asia (Area), **Smallest:** Australia.
- **Most countries:** Africa (54).
- **Two smallest continents:** Australia & Europe.
- **Highest plains proportion:** Europe.
- **Highest mean elevation:** Antarctica.
- **Guyana is NOT in Africa**, but in **South America** (English official language).
- **Geographical features:**
 - Aconcagua → South America
 - Meseta Plateau → Europe (Spain)
 - Serengeti Plains → Africa (Tanzania)
 - Gibson Desert → Australia
- **Atlas Mountains** → NW Africa; **Abyssinian Plateau** → East Africa; **Guiana Highlands** → N. South America; **Okavango Basin** → SW Africa.

MOUNTAINS

DEFINITION & CHARACTERISTICS

- **Mountain Range:** Series of mountains in linear arrangement, connected by high ground.
- **Difference with Plateau:**
 - Mountain → elevated, pointed summit.
 - Plateau → elevated, flat top.
- **Classification:** Elevation ≥ **1000 m** = mountain; <1000 m = hill.
- **Coverage:** Occupy ~25% of Earth's land but support only **12% of population**.
- **Nickname:** **British Columbia** (Canada) = *Sea of Mountains* (Molyneux St. John, 1876).



TYPES OF MOUNTAINS

1. FOLD MOUNTAINS (FORMED BY FOLDING OF CRUST LAYERS)

- **Andes** (South America) – **longest range in world** (7000–8900 km); highest peak = **Aconcagua (6962 m)**.
- **Rockies** (North America) – length ~4800 km; highest peak = **Mount Elbert (4401 m)**.
- **Himalayas** (Asia) – ~2500 km; highest peak = **Mount Everest (8848.86 m, Sagarmatha/Chomolungma)**.

- **Southern Alps** (New Zealand) – highest = **Aoraki/Mt. Cook (3724 m)**.
- **Alps** (Europe) – young fold mountains; highest = **Mont Blanc (4807 m)**.
- **Atlas** (Africa) – ~2500 km, in Morocco, Algeria, Tunisia.
- **Pyrenees** – natural boundary of Spain & France.
- **Apennines** (Italy) – length 1350 km; highest = **Corno Grande (2914 m)**.
- **Dinaric Alps** – Italy → Balkans.
- **Jura** – France & Switzerland.
- **Altai** – Asia.

2. BLOCK MOUNTAINS (FAULTING/UPLIFT)

- **Sierra Nevada** (USA) – largest block mountain.
- **Black Forest** (Germany) – highest = **Feldberg (1493 m)**.
- **Vosges** (France).
- **Salt Range** (Pakistan).

3. DOME MOUNTAINS (UPLIFTED BY MAGMA, NO ERUPTION)

- Examples: **Cincinnati Arch (USA)**, **Black Hills, Bighorns**.
- **Fujiyama** (Japan), **Cotopaxi** (Ecuador) – volcanic domes.

4. RESIDUAL MOUNTAINS (EROSION/DENUATION)

- **Aravallis** (India), **Appalachians** (USA), **Pennines** (UK/Europe).

5. MIXED/COMPLEX MOUNTAINS

- **Sierra Nevada, Anaconda** (USA) – complex origins.

MAJOR PEAKS BY CONTINENT

- **Asia** – Himalayas (Everest – 8848.86 m).
- **Africa** – Kilimanjaro (5895 m, Tanzania).
- **Europe** – Elbrus (5642 m, Russia, extinct volcano).
- **North America** – Denali/McKinley (6190 m, Alaska).
- **South America** – Aconcagua (6962 m, Argentina).
- **Australia** – Kosciuszko (2228 m, NSW).
- **Antarctica** – Vinson Massif (4892 m).

OTHER IMPORTANT RANGES

- **Allegheny Mountains** – USA (Appalachians).
- **Cantabrian Mountains** – Spain.
- **Mackenzie Mountains** – Canada.
- **Elburz Mountains** – Iran.
- **Drakensberg** – South Africa & Lesotho; highest = **Thabana Ntlenyana (3482 m)**.
- **Golan Heights** – SW Syria, basaltic plateau, annexed by Israel (1981).
- **Khasi, Jaintia, Garo Hills** – Meghalaya (India).
- **Arakan Yoma** – Myanmar (boundary with India); highest = **Natma Taung/Mt. Victoria**.
- **Ural Mountains** – Old folded mountains, boundary of Europe-Asia.

MOUNTAIN FORMATION (GEOLOGICAL ERAS)

- **Tertiary (Alpine Orogeny)** – Alps, Andes, Himalayas, Rockies, Atlas.

- **Caledonian** – Appalachians.
- **Pre-Cambrian** – Aravallis.
- **Hercynian** – Black Forest, Vosges, Tien Shan, Nan Shan.

FACTS

- **Longest range:** Andes (7000–8900 km).
- **Young Folded Mountains:** Himalayas, Andes, Alps, Rockies.
- **Old Mountains:** Appalachians, Aravallis, Pennines, Urals.
- **Highest Peaks (Young Folded)** – mostly in Himalayas.
- **Mountains & Locations:**
 - Atlas → Africa (Morocco, Algeria, Tunisia).
 - Pyrenees → France-Spain border.
 - Apennines → Italy.
 - Dinaric Alps → Balkans.
 - Drakensberg → South Africa.
- **Mountains & Countries (Match-the-pair):**
 - Kilimanjaro → Tanzania
 - Toubkal → Morocco
 - Stanley → Uganda
 - Hoggar → Algeria
- **Residual Mountain Example:** Vindhya (India).
- **Volcanic Peak:** Fujiyama (Japan).
- **White Mountains:** USA (New Hampshire, Maine).
- **Black Mountains:** USA (North Carolina, Appalachians).

THE PLATEAUS

DEFINITION & CHARACTERISTICS

- **Plateau (Tableland/High plain):** Elevated flat terrain, raised above surrounding area, often with steep sides.
- **Elevation:** Second only to mountains; **area coverage:** second only to plains.
- **Global Coverage:** Found on every continent.

TYPES OF PLATEAUS & EXAMPLES

1. Glacial Plateau

- Formed by **erosional & depositional work of glaciers**.
- Examples: **Prussian Plateau (Germany), Marg (Kashmir, India).**

2. Fluvial Plateau

- Formed by continuous **river deposition**.
- Examples: **Vindhya Plateau, Shillong Plateau (India), Shan Plateau (Myanmar).**

3. Intermontane Plateau (highest, enclosed by mountains)

- Located between mountain ranges, surrounded on all sides.
- **Tibetan Plateau (Qinghai–Tibet Plateau):**
 - Area: ~2.5 million km².
 - **Average elevation: 4500 m (world’s highest plateau).**
 - Includes **Chang Tang (Tibet–Xinjiang–Ladakh).**
- **Pamir Plateau (Central Asia):**
 - Called “**Roof of the World**” (originally for Pamir knot, later for Tibet too).

- Junction of Himalayas, Tien Shan, Karakoram, Kunlun, Hindu Kush, Suleiman ranges.

- Other examples: **Bolivia, Peru, Colombia (South America); Mexican Plateau.**

4. Piedmont Plateau (at foothills)

- Name origin: *Piedmont* = “foothill” (Latin *Pedemontium*).
- Examples: **Piedmont Plateau (Eastern USA), Patagonia Plateau (South America)** – rich in minerals.

5. Domed Plateau

- Formed by **uplift of land into a dome**.
- Example: **Ozark Plateau (USA).**

6. Continental Plateau (part of continents, broad uplift)

- Examples: **Indian Peninsula, Arabian Plateau, Australia Plateau, South African Plateau, Iberian Peninsula (Spain & Portugal).**
- **Meseta Plateau (Spain):**
 - Avg. elevation: **610–760 m**.
 - **Madrid (Spain’s capital)** located on it.

7. Coastal Plateau

- Near sea coasts.
- Example: **Coromandel Coast (India).**

8. Volcanic Plateau

- Formed by **lava or pyroclastic flows**.
- **Lava Plateau:** Basaltic lava, repeated eruptions.
- **Pyroclastic Plateau (Ignimbrite):** From volcanic ash, tuff, tephra.
- Examples: **Deccan Plateau (India), Plateaus of NZ, South Africa, Argentina, Brazil, Western USA, Mexico, France, Siberia.**

FACTS

- **World’s largest plateau: Tibetan Plateau (avg. 4500 m elevation).**
- **Roof of the World:** Pamir Plateau (also used for Tibetan).
- **Patagonia Plateau (Argentina)** → storehouse of minerals.
- **Colorado Plateau (USA)** → largest plateau in North America (150,580 sq. miles).
- **Telegraph Plateau** → Part of **North Atlantic Ridge** (between Newfoundland & Ireland).
- **Oceanic Plateaus** examples:
 - Telegraph Plateau → Atlantic
 - Cocoa Ridge → Pacific
 - Walvis Ridge → Atlantic
 - Agulhas Basin → Indian Ocean
- **Tibetan Plateau:** Avg. elevation **4500 m+** → highest in world.
- **Patagonia Plateau:** Storehouse of minerals (South America).
- **Madrid:** Located on Meseta Plateau (Spain).
- **Roof of the World:** Pamir Plateau (Central Asia).
- **Largest Plateau (World):** Tibetan; **Largest in North America:** Colorado Plateau.
- **Telegraph Plateau:** In **Atlantic Ocean**, not Indian.
- **Incorrect matches:**
 - Telegraph Plateau ≠ Indian Ocean (it’s in Atlantic).

VALLEY

DEFINITION & TYPES

- **Valley:** Low area between hills/mountains, usually with a **river flowing through**.
- In geology → **long depression with limited width**.
- **Rift Valley:**
 - Formed at **divergent plate boundaries** (crustal extension & faulting).
 - Later deepened by erosion.
 - Example: **Death Valley (California, USA)**.

FAMOUS RIFT VALLEYS

- **Death Valley (USA)**
 - Location: California, Mojave Desert.
 - Lowest point in **North America: -86 m** (2nd lowest in Western Hemisphere after **Laguna del Carbón, Argentina -105 m**).
 - **Hottest & driest place in USA**; temp up to **56.7°C**.
 - Contains **Devil's Golf Course** = salt pan (remnant of dried Holocene lake).
- **Other Rift Valleys:** East African Rift (Kenya, Ethiopia, Tanzania), Jordan Rift (Dead Sea).

OTHER IMPORTANT VALLEYS

- **Panjshir Valley (Afghanistan)** – north of Kabul, near Hindu Kush; historic **battle-ground in Soviet war**.
- **Taylor Valley (Antarctica)** – discovered by **Robert Falcon Scott (1903)**, called *Valley of the Dead*.
- **Chumbi Valley** – Border of India, Bhutan, Tibet.
- **Kathmandu Valley (Nepal)** – cultural hub.
- **Swat Valley (Pakistan)** – scenic valley.
- **Po Valley (Italy)** – fertile agricultural region.
- **Sacred Valley (Peru)** – near Andes, Inca civilization.
- **Ebro Valley (Spain)**.
- **Loire Valley (France)** – famous for vineyards/castles.
- **Rabbit, Pleasant, Grand Valley (USA)**.
- **Blackwood & King Valley (Australia)**.
- **Bada Valley (Indonesia)**.
- **Humber Valley (Canada)**.
- **Wolgan Valley (Australia)**.

SPECIAL VALLEYS & REGIONS

- **Silicon Valley (USA)** – in **California**, hub of **computer chip & tech industries**.
- **Valley of the Kings (Egypt)** – *Wadi el-Muluk*, ancient royal tombs.
- **Blind Valley** – found in **Karst regions**; abrupt closed valley.
- **Sinkholes (Dolines)** – also Karst landforms.

BASINS & RELATED LANDFORMS

- **Great Artesian Basin (Australia)** – world's **largest artesian basin** (~1.7 million km²); vital water source.
- **Great Basin (USA)** – largest endorheic watershed in North America; between **Rockies & Sierra Nevada**; bordered north by Columbia Plateau, south by Mojave & Sonoran deserts.

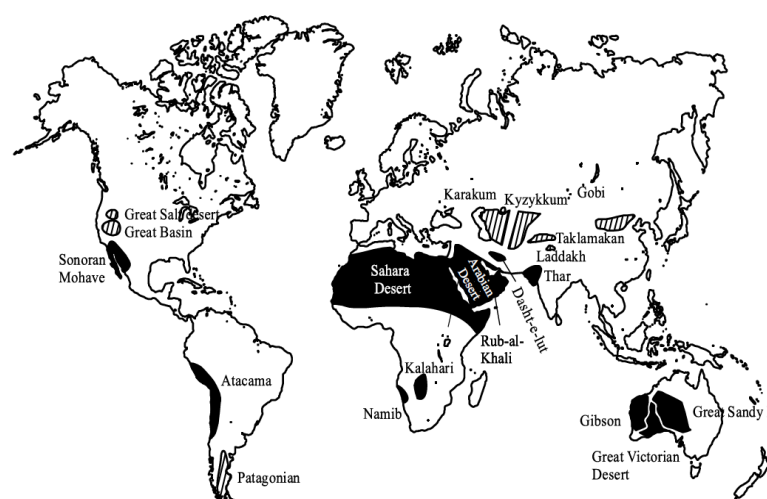
- **Columbia Plateau (USA)** – north of Great Basin.

FACTS

- **Death Valley:** Rift Valley, in California; hottest, driest, lowest in N. America (-86 m).
- **Devil's Golf Course:** Salt pan in Death Valley.
- **Panjshir Valley:** Afghanistan (north of Kabul).
- **Taylor Valley:** Antarctica.
- **Silicon Valley:** California (computer industry hub).
- **Great Artesian Basin:** Australia.
- **Great Basin:** USA (endorheic).
- **Valley of the Kings:** Egypt.
- **Blind Valley & Sinkholes:** Karst topography.

THE ARID REGIONS / DESERTS

Major Deserts of the World



DESERT – DEFINITION & DISTRIBUTION

- **Definition:** Barren area with **<25 cm annual rainfall**, hostile for plants & animals.
- **Global Coverage:** ~1/3rd of Earth's land is arid/semi-arid.
- **Location:**
 - Hot deserts mostly on **western margins of continents** (15°–30° N & S).
 - Associated with **trade winds, cold ocean currents, subtropical high-pressure belts**.
- **Renewable Energy Potential:** Dr. Gerhard Knies → World deserts receive in **6 hrs more solar energy than annual human consumption**.

MAJOR HOT DESERTS

AFRICA

- **Sahara** – Largest non-polar desert (~3.6 million sq. mi.); spans 11 countries.
- **Kalahari** – Semi-arid, ~9,30,000 km²; Botswana, Namibia, South Africa.
- **Namib** – Coastal desert along SW Africa (influenced by cold Benguela current).

ASIA

- **Thar Desert (India-Pakistan)** – Largest in South Asia; **densest populated desert (83/km²)**.
- **Dasht-e-Lut (Iran)** – World's **hottest desert (70.7°C, NASA 2005)**.

INTERNATIONAL BORDER LINES

CONCEPT

- **International Border Lines** = Artificial or natural boundaries separating countries.
- Can be based on **latitudes, rivers, mountain ranges, or man-made fortifications**.

IMPORTANT INTERNATIONAL BORDERS

India

- **McMahon Line** – *India–China border*, drawn in **1914 Shimla Accord** (between Britain & Tibet, rejected by China).
- **Radcliffe Line** – *India–Pakistan & India–Bangladesh border* (drawn in **1947** by Sir Cyril Radcliffe).
- **Durand Line** – *India–Afghanistan & Pakistan–Afghanistan border* (drawn **1893** between Sir Mortimer Durand & Afghan Amir Abdur Rahman Khan).

Europe

- **Maginot Line** – France–Germany border (built **1930–35** after WWI, named after French Defence Minister *André Maginot*).
- **Alpine Line (Little Maginot Line)** – France–Italy border.
- **Hindenburg Line** – German defensive line (WWI, 1916–17) → Germany–Belgium–France border.
- **Oder-Neisse Line** – Present border between *Germany & Poland*.
- **Mannerheim Line** – *Finland–Russia border* (WWII).

Asia

- **38th Parallel (38°N)** – Divides *North Korea & South Korea*.
- **49th Parallel (49°N)** – Forms much of *USA–Canada border*.
- **Lake Victoria (Africa, but important)** – Border between *Uganda–Tanzania, Uganda–Kenya, Kenya–Tanzania*.
- **Rio Grande River** – Border between *USA & Mexico*.

FACTS

- **Rio Grande** → USA–Mexico border.
- **Rhine River** → Forms *France–Germany border*.
- **Lake Victoria (Nyanza)** → Second largest freshwater lake in world; border between *Kenya, Uganda, Tanzania*.
- **Maginot Line** → France–Germany.
- **Radcliffe Line** → India–Pakistan, India–Bangladesh.
- **Durand Line** → India–Afghanistan, Pakistan–Afghanistan.
- **McMahon Line** → India–China (not recognized by China).
- **38th Parallel (38°N)** → North & South Korea.
- **49th Parallel (49°N)** → USA–Canada border.
- **Oder-Neisse Line** → Germany–Poland.
- **Hindenburg Line** → Germany–Belgium–France (WWI).

CAPITAL CITIES OF THE WORLD

KEY CONCEPTS

- **Capital City** = administrative, political, or cultural hub of a country.
- Some countries have **former capitals** or **multiple capitals** (official vs de facto).

SPECIAL CASES & FORMER CAPITALS

- **Australia** → Inaugural capital: *Melbourne*; Present: *Canberra*.
- **New Zealand** → Former: *Auckland*; Present: *Wellington*.
- **Japan** → Former: *Kyoto*; Present: *Tokyo*.
- **Pakistan** → Former: *Karachi*; Present: *Islamabad*.
- **Germany** → Former West German capital: *Bonn*; Present: *Berlin*.
- **Myanmar (Burma)** → Former: *Yangon (Rangoon)*; Since 2005, *Nay Pyi Taw*.

CAPITALS WITH INTERNATIONAL IMPORTANCE

- **Vienna (Austria)** → HQ of *International Atomic Energy Agency (IAEA)*.
- **New York (USA)** → *Not a capital*; Washington D.C. is the capital.
- **The Hague (Netherlands)** → Seat of Government, but *Amsterdam* is official capital.
- **Yamoussoukro (Côte d'Ivoire/Ivory Coast)** → Official capital; *Abidjan* functions as de facto capital.

FACTS

- **Slovenia** → Capital: *Ljubljana* (not Bratislava, which is Slovakia).
- **Nigeria** → Capital: *Abuja* (not Rabat → Morocco).
- **North Korea** → Capital: *Pyongyang* (not Seoul, which is South Korea).
- **Ethiopia** → Capital: *Addis Ababa* (not Lusaka → Zambia).
- **Maldives** → Capital: *Male* (not Malabo → Equatorial Guinea).
- **Dominica** → Capital: *Roseau* (not Santo Domingo → Dominican Republic).

SOME MUST-MEMORIZE CAPITALS

ASIA

- **Afghanistan** → Kabul
- **Kazakhstan** → Nur Sultan (Astana)
- **Kyrgyzstan** → Bishkek
- **Uzbekistan** → Tashkent
- **Tajikistan** → Dushanbe
- **Turkmenistan** → Ashgabat
- **Cyprus** → Nicosia
- **Lebanon** → Beirut
- **Iran** → Tehran
- **Iraq** → Baghdad

- **Israel** → Jerusalem

EUROPE

- **Slovenia** → Ljubljana
- **Slovakia** → Bratislava
- **Hungary** → Budapest
- **Latvia** → Riga
- **Lithuania** → Vilnius
- **Norway** → Oslo
- **Sweden** → Stockholm
- **Finland** → Helsinki
- **Switzerland** → Bern

AFRICA

- **Nigeria** → Abuja
- **Kenya** → Nairobi
- **Namibia** → Windhoek
- **Ghana** → Accra
- **Zambia** → Lusaka
- **Rwanda** → Kigali
- **Sudan** → Khartoum
- **DR Congo** → Kinshasa
- **Cape Verde** → Praia

AMERICAS

- **Brazil** → Brasilia (not Rio de Janeiro)
- **Colombia** → Bogotá
- **Venezuela** → Caracas
- **Peru** → Lima
- **Paraguay** → Asunción
- **Guyana** → Georgetown
- **Costa Rica** → San José
- **Belize** → Belmopan (not Managua → Nicaragua)

OCEANIA

- **Australia** → Canberra
- **New Zealand** → Wellington
- **Fiji** → Suva
- **Papua New Guinea** → Port Moresby

FACTS

- **Sydney & New York** → Famous cities, but *not capitals*.
- **Pisa (Italy)** → Famous for Leaning Tower; not a national capital (Rome is capital).
- **Christchurch (NZ)** → Not a capital; Wellington is the capital.
- **Brisbane (Australia)** → Not a capital; Canberra is the capital.
- **Bolivia** → La Paz (administrative) & Sucre (constitutional).

WORLD TOWNS

MEGA CITIES & LOCATION

- Most **mega cities** are in **coastal areas** because they are gateways to **international ocean routes**.

CITIES ON LAKES (NORTH AMERICA)

- **Erie** → Detroit

- **Michigan** → Gary
- **Ontario** → Hamilton
- **Superior** → Duluth

CENTRAL ASIA (CAPITALS MAP QS)

- **Ashgabat** → Turkmenistan
- **Tashkent** → Uzbekistan
- **Bishkek** → Kyrgyzstan
- **Dushanbe** → Tajikistan

CAPITALS OF SPECIAL TERRITORIES

- **Palestine** → Ramallah
- **Kosovo** → Pristina
- **Turkish Cyprus** → Nicosia
- **Taiwan** → Taipei

FAMOUS CITY IDENTIFICATIONS

- **Darwin** → North Australia
- **Kuala Lumpur** → Malaysia
- **Nairobi** → Kenya
- **Lagos** → Nigeria
- **Jaffna** → Northern Sri Lanka

ALASKA

- Belongs to the **USA**, purchased from **Russia** (1867).

SILICON VALLEY (CALIFORNIA, USA)

- Includes: Campbell, Palo Alto, Santa Clara

SOUTH AFRICA (CAPITALS)

- **Cape Town** → Legislative capital (Parliament seat)
- **Pretoria** → Administrative capital
- **Bloemfontein** → Judicial capital

PAKISTAN (NORTH → SOUTH)

- Peshawar → Islamabad → Gujranwala → Multan

AFGHANISTAN

- **Kandahar** → Southern Afghanistan

EUROPE (WEST → EAST)

- Lisbon → London → Frankfurt → Beirut

HIGH-ALTITUDE CITIES

- **Lhasa (Tibet)** → 3650 m (highest)
- Thimphu (Bhutan) → ~2300 m
- Gartola (Nepal) → 1733 m
- Kathmandu (Nepal) → 1350 m

SOUTH-EAST ASIA (SOUTH → NORTH)

- Jakarta (Indonesia, ~6°S)
- Singapore (~1°N)
- Bangkok (Thailand, ~13°N)
- Hanoi (Vietnam, ~21°N)

- Terra Rossa → Red from Limestone
- Chestnut → Dry = Low Humus

THE MAJOR TRIBES OF THE WORLD

GENERAL

- **Tribes** → groups based on common **language, culture & customs**.
- **Nomadic tribes** → move in search of pasture, esp. in **arid regions (Africa, Central Asia, South America)**.
- **Transhumance** → seasonal migration (valley ↔ mountains).
 - *Examples (India):* Gaddi, Bakarwal, Bhotia.

IMPORTANT TRIBES BY REGION

AFRICA

- **Pygmies** (Congo Basin) → hunters, gatherers; live in tree huts (*Mongulu*); *short stature, Negrito*.
- **Maasai** (Kenya, Tanzania, Uganda) → cattle herders; huts = *Kraal*.
- **Bushmen** (Kalahari, Botswana, Namibia, S. Africa) → hunters & gatherers, eat termites = “rice of Bushmen”.
- **Zulu** (South Africa) → agriculture, cattle rearing; famous *Battle of Isandlwana (1879)*.
- **Berbers** (Morocco, N. Africa).
- **Nuba** (Sudan).
- **Fulani** (West Africa).
- **Bantu** (East, Central, Southern Africa).
- **Ivory Coast** → hot, humid, dense forests.

ASIA

- **Semang** (Malaysia, Negrito) → hunters.
- **Sakai** (Malaysia, Indonesia) → use blowpipe.
- **Bedouins** (Arab deserts, esp. Saudi Arabia) → nomads, herders.
- **Kirghiz** (Central Asia) → cattle rearing, seasonal migration; huts = *Yurt*.
- **Yukaghir** (NE Siberia, Russia).
- **Lapps/Sami** (Fenno-Scandinavia: Norway, Sweden, Finland, Kola Peninsula).
- **Ainu** (Japan).

THE AMERICAS

- **Eskimos/Inuit** (Alaska, Canada, Greenland, Russia) → Mongoloid race, hunters; use **Harpoon, Kayak (hunting boat), Umiak (transport), Sledge (dog-cart)**.
- **Yupik** (Eastern Russia, Western Alaska).
- **Maya** (Central America: Mexico, Guatemala, Honduras).
- **Bora** (Western Amazon Basin: Brazil, Peru, Colombia).
- **Red Indians** (North America, Mongoloid origin).

PACIFIC / OCEANIA

- **Papua** (Papua New Guinea) → agriculture, cattle rearing.
- **Maori** (New Zealand, Polynesian) → agriculture, gathering.

- **Aeta** (Philippines).

ANTHROPOLOGY & RACES

- **Caucasoid** (largest race today) → straight/curly hair, blue/brown eyes. Subtypes: Nordic, Alpine, Mediterranean.
- **Mongoloid** → Central & East Asia, yellowish-brown skin, small forehead.
- **Negroid** → Africa, dark brown to black.
- **Australoid** → S. India, Australia.
- **Cro-Magnon** → most modern prehistoric man (40,000–10,000 yrs ago, Dordogne Valley, France).
- **Archaeopteryx** → first bird (~150 million years ago).
- **Africa = “Cradle of Mankind”**.

FACTS

1. **Nomadic herding** → common in **dry regions**.
2. **Transhumance** → seasonal migration (Bhotiya, Bakarwal).
3. **Cro-Magnon** → most recent prehistoric man.
4. **First bird** → Archaeopteryx (150 mya).
5. **Cradle of mankind** → Africa.
6. **Largest racial group** → Caucasoid.
7. **Pygmies** → Congo Basin, Equatorial forests.
8. **Eskimo/Inuit** → Canada, Alaska, Greenland (igloos, sledge, kayak).
9. **Bushman** → Kalahari Desert.
10. **Maasai** → East Africa (Kenya, Tanzania).
11. **Maori** → New Zealand.
12. **Sami (Lapps)** → Scandinavia.
13. **Incorrect matches often asked** → Maasai (not Zambia, but East Africa), Semang (Malaysia, not Indonesia), Bantu (not Sahara, but E/S Africa).

THE LANGUAGES OF THE WORLD

LINGUISTIC CENSUS

- **L-1** = Primary language / Mother tongue.
- **L-2** = People who speak the language as a **second language**.
- **Total speakers** = L-1 + L-2.

TOP 8 MOST SPOKEN LANGUAGES (ETHNOLOGUE, 2022)

Rank	Language	L-1 (Mother tongue)	L-2 (Second language)	Total Speakers
1	English	379.7 M	1076.7 M	1.45 B
2	Mandarin	939.2 M	199.0 M	1.14 B
3	Hindi	344.7 M	264.8 M	609 M
4	Spanish	485.1 M	74.0 M	559 M
5	French	80.8 M	229.0 M	310 M
6	Bangla	233.8 M	39.0 M	273 M
7	Arabic	—	—	274 M

- **By L-1 (mother tongue only)** → **Mandarin = #1**.
- **By total speakers (L1 + L2)** → **English = #1**.

SPECIAL LANGUAGES

- **Esperanto** → an **artificial international auxiliary language**, created by *Dr. Ludwik Zamenhof*, used by 1–2 million people.
- **Urdu** → spoken in Pakistan.
- **Burmese** → spoken in Myanmar.

OFFICIAL LANGUAGES IN SELECTED COUNTRIES

- **Mauritius** → English & French (official). Also Hindi, Urdu, Hakka, Bhojpuri, Creole.
- **Singapore** → Malay (National), **English, Tamil, Mandarin** (official).
- **Chile, Colombia, Cuba** → Spanish.
- **Pakistan** → Urdu & English (official), Punjabi widely spoken.
- **Indonesia** → Bahasa Indonesia (official).
- **Thailand** → Thai (NOT Bahasa).

AGRICULTURE & LIVESTOCK

RICE

- **Climate:** Wet tropical, needs humidity & standing water.
- **Season:** Kharif crop (harvest in Nov–Dec); in Java (Indonesia) → harvested in Apr–May.
- **Research:** *International Rice Gene Bank* → Los Banos, Philippines (IRRI, est. 1960).
- **Hybrid rice:** Widely cultivated in China (≈50% of rice area).
- **FAO 2022 Data:**
 - **Top producers:**
 1. China → 208.49 MT
 2. India → 196.24 MT
 3. Bangladesh → 57.18 MT
 4. Indonesia → 54.74 MT
 5. Vietnam → 42.67 MT
 - **Area under cultivation:**
 - India → 46.40 M ha (**largest area**)
 - China → 29.68 M ha
 - **Productivity:** Australia (110,545 hg/ha) > Uruguay > China (70,796 hg/ha) > Japan (69,208 hg/ha) > India (42,294 hg/ha).

FACTS

1. Leading producer of rice → **China**
2. India's rank in production → **2nd**
3. Largest area → **India**
4. Hybrid rice cultivation → **China**
5. Rice Gene Bank → **Philippines**
6. Country earning ⅓rd of exports from rice trade → **Myanmar** (when asked; today top = USA, Brazil)

WHEAT

- **Climate:** Temperate crop.
 - Germination → ~10°C
 - Grain development → 15–20°C
 - Rainfall needed → 20–70 cm.
- **Latitude:** Up to 60°N, 40°S.

FAO 2022 DATA:

- **Top producers:**
 1. China → 137.72 MT

2. India → 107.74 MT
 3. Russia → 104.23 MT
 4. USA → 44.90 MT
- Largest wheat producer → **China**.

FACTS

1. Leading producer of both wheat & rice → **China**
2. Order (descending): **China > India > Russia > USA**
3. Largest wheat producer → **China**

SUGARCANE, SUGAR BEET & COTTON

SUGAR CROPS

SUGARCANE

- **Climate:** Sub-tropical & monsoonal.
 - Temp: 21–27 °C
 - Rainfall: 75–150 cm.

FAO 2022 TOP PRODUCERS:

3. Brazil → **724.42 MT**
4. India → **439.42 MT**
5. China → 103.38 MT
6. Thailand → 92.09 MT
7. Pakistan → 87.98 MT

FACTS

- Largest producer → **Brazil**
- 2nd largest → **India**

SUGAR BEET

- **Climate:** Temperate crop.
- **FAO 2022 Top Producers:**
 - Russia
 - France
 - (followed by USA, Germany, Turkiye)
- **Special:** Ukraine → produces *all sugar* from sugar beet.

FACTS

- Two largest producers → **Russia & France**
- Country with only sugar beet sugar → **Ukraine**

COTTON

- **Climate:** Tropical & Sub-tropical.
- **Quality depends on:** fibre length, shine, etc.

COTTON SEED (FAO 2021)

1. India → **11.62 MT**
2. China → 9.60 MT
3. USA → 4.82 MT
4. Brazil → 3.43 MT
5. Pakistan → 2.67 MT

COTTON LINT (FAO 2021)

1. India → **5.99 MT**
2. China → 5.73 MT
3. USA → 3.81 MT
4. Brazil → 2.22 MT
5. Pakistan → 1.41 MT

PRODUCTIVITY (FAO 2022)

- **Composition:**
 - General: Cu 60%, Ni 20%, Zn 20%
 - NCERT: Cu (25–30%), Zn (25–30%), Ni (40–50%)

FACTS

1. **Largest producer of silver (world): Mexico**
2. **Mexico is the largest producer of: Silver**
3. **Does not contain silver: German Silver** (only Cu + Zn + Ni).

ALUMINIUM (AL)

KEY FACTS

- **Whitish, light-weight, ductile metal**
- **Bauxite = chief ore of Aluminium**
- Uses → Aircraft, construction, electrical wires, utensils, packaging (foil, cans).

MAJOR BAUXITE REGIONS

- **Australia** → Cape York Peninsula, Weipa
- **USA** → Saline County (Arkansas)
- **Russia** → Kola Peninsula
- Also in India (Odisha, Jharkhand, Chhattisgarh, Gujarat).

TOP 5 ALUMINIUM PRODUCERS (PRIMARY, 2021 – WORLD MINERAL PRODUCTION)

1. **China** – 38,502.60 ('000 MT)
2. **Russia** – 3,640.00 ('000 MT)
3. **India** – 3,583.80 ('000 MT)
4. **Canada** – 3,157.76 ('000 MT)
5. **UAE** – 2,540.00 ('000 MT)

TIN (SN)

KEY FACTS

- About **80% of world tin** → found in **placer deposits** (river beds, valleys, sea floors).
- Important **Southeast Asia concentration** → “Tin Belt of SE Asia”.
- Used in → **Alloys (bronze, solder), coating, tin plating, packaging (cans).**

MAJOR TIN REGIONS

- **China** → Yunnan Plateau
- **Indonesia** → Sumatra (Bangka & Belitung islands)
- **Malaysia** → Kinta Valley (Perak), Kuala Lumpur, Pahang
- **South America** → Eastern Andes belt (800 km × 96 km, esp. Bolivia, Peru)

TOP TIN PRODUCERS (MINE PRODUCTION, 2021 – WMP)

1. **China** – 80,000 tonnes
2. **Indonesia** – 34,466 tonnes
3. **Myanmar** – 32,000 tonnes
4. **Peru** – 26,995 tonnes

PETROLEUM & NATURAL GAS

PETROLEUM & NATURAL GAS

- Petroleum = liquid hydrocarbon found in **sedimentary rocks** (sandstone, limestone).
- First **commercial oil well** → Romania (1857).
- First **U.S. oil well** → Titusville, Pennsylvania (1859).
- Major share of world reserves → **Middle East (48.3%)**.
- Venezuela → exports ~87% revenue from oil.
- USA = **largest producer (2021)**.

MAJOR OIL PRODUCERS (2021 – WMP)

1. **USA** – 748.13 MT
2. **Russia** – 536.40 MT
3. **Saudi Arabia** – 515.00 MT
4. **Canada** – 234.50 MT
5. **Iraq** – 200.80 MT

TOP OIL RESERVES (IMYB 2021)

1. **Venezuela** – 48.0 Bt
2. **Saudi Arabia** – 40.9 Bt
3. **Canada** – 27.1 Bt
4. **Iran** – 21.7 Bt
5. **Iraq** – 19.6 Bt

NATURAL GAS RESERVES (IMYB 2021)

1. Russia
2. Iran
3. Qatar
4. Turkmenistan
5. USA

Top Producers (2020): USA > Russia > Iran > China > Qatar

IMPORTANT REGIONS

- **Asia (Middle East):** Saudi Arabia, Iran, Iraq, UAE, Kuwait, Oman, Qatar
- **SE Asia:** Indonesia (Sumatra, Java, Kalimantan) → #1 in region
- **China:** Sichuan, Shanxi, Yumen, Saidum
- **USA:** Texas, New Mexico, North Dakota (Alaska also)
- **Canada:** Alberta (Leduc), Manitoba, British Columbia
- **Russia/CIS:** Volga–Ural, Baku (Azerbaijan), Caspian Sea belt
- **South America:** Venezuela, Brazil, Colombia

FACTS

- **Gasohol** = 90% Gasoline + 10% Ethanol (USA largest producer & consumer).
- **Brent Crude Oil** = benchmark, from **North Sea**, sulphur content ~0.37%.
- **Biofuels:**
 - Gasohol → Sugarcane, Maize
 - Biodiesel → Jatropha oil
- **Peace Pipeline** → Iran–Pakistan gas pipeline.
- **Ras Tanura Refinery** → Saudi Arabia (Persian Gulf coast).
- 1. First oil well → Romania (1857).
- 2. Largest oil producer (2021) → USA.

3. Largest oil reserves → Venezuela.
4. Largest SE Asian producer → Indonesia.
5. Largest exporter to India (2017–18) → Iraq.
6. **Brent Crude** → North Sea, benchmark oil.
7. Peace Pipeline → Iran–Pakistan.
8. Pegu Yoma (Myanmar) → Mineral oil.

URANIUM & THORIUM

- **Discovered (1789):** Martin Klaproth.
- **Main ore: Pitchblende.**
- **Atomic minerals:** Uranium & Thorium → key for nuclear power.
- **Radium** → extracted from Uranium ore (Pitchblende).

TOP URANIUM PRODUCERS (2022 – WNA)

1. **Kazakhstan**
2. Canada
3. Namibia
4. Australia
5. Uzbekistan
- India = 9th**

TOP URANIUM RESERVES (2021 – WNA/USGS)

1. **Australia** (Olympic Dam, Ranger)
2. Kazakhstan
3. Canada
4. Russia
5. Namibia

IMPORTANT MINING REGIONS

- **Australia:** Olympic Dam, Mary Kathleen, Radium Hill
- **Canada:** Cigar Lake, Port Radium, Elliot Lake, Uranium City (Lake Athabasca, Saskatchewan)
- **USA:** Colorado Plateau
- **Congo (DRC):** Shinkolobwe mine (Katanga)
- **Other reserves:** Niger, Namibia, Uzbekistan

THORIUM

- Max known reserves → **Kerala coast (India) & Espirito Santo (Brazil).**
- Also found in **USA (SE & W mountains), Nigeria, South Africa, Russia, Indonesia (Sumatra).**

FACTS

1. Largest producer of Uranium → **Kazakhstan.**
2. Largest reserves of Uranium → **Australia.**
3. Largest Uranium mine (single place, production) → **Cigar Lake (Canada).**
4. Uranium City → **Canada (Lake Athabasca, Saskatchewan).**
5. Radium → from **Pitchblende** ore.
6. Congo (Katanga/Shinkolobwe) → Uranium deposits.
7. Thorium max reserves → **India (Kerala sands) & Brazil (Espirito Santo).**
8. Before 2009 → **Canada** was #1 producer (replaced by Kazakhstan).

BAUXITE (ALUMINIUM ORE)

- **Reserves (USGS 2023):** Guinea > Vietnam > Australia > Brazil > Jamaica
- **Production (2021):** Australia (#1, ~30%) > Guinea > China > Brazil > Indonesia > India (#6)
- **Key regions:** Cape York & Weipa (Australia), Saline County (USA), Cola Peninsula (Russia).

TIN (CASSITERITE ORE)

- ~80% from **placer deposits** (riverbeds, valleys, sea floors).
- **Main areas:** SE Asia – Yunnan (China), Bangka & Belitung (Indonesia), Kinta Valley (Malaysia), Andes (Peru/Bolivia).
- **Top producers (2021):** China > Indonesia > Myanmar > Peru.

DIAMOND

- **Ores:** Kimberlite pipes, Conglomerates, Alluvial gravels.
- **Reserves (USGS 2023):** Russia > Botswana > DRC.
- **Production (2021):** Russia > Botswana > Canada > DRC > South Africa.
- **Famous centres:** Kimberley (S. Africa), Catoca (Angola), Mirny (Russia).

TITANIUM

- Strong, light → used in aerospace, steel substitute.
- **Largest producer (2022):** China.

NITRATES (SALTPETRE – KNO₃)

- Found in **Chile & Peru** (Atacama Desert).
- Largest producer → **Chile.**

RARE EARTH METALS (17 ELEMENTS)

- Used in **mobiles, EV batteries, wind turbines, missiles.**
- **Production (2022):** China (~70%) > USA > Myanmar > Australia.
- **Reserves (India):** ~6.9 MT (Andhra, Bihar, Kerala, Odisha, TN, WB).
- China cut exports in **2010**, causing global supply concerns.

CAMPHOR

1. **Cinnamomum camphora tree** (China, Japan).
2. **Dryobalanops aromatica tree** (Borneo, Sumatra).
3. **Blumea spp.** (India).

OTHER KEY MINERALS

- **Port Radium (Canada):** Uranium.
- **Bingham Canyon, Utah (USA):** Copper.
- **Baku (Azerbaijan):** Petroleum.
- **Mesabi Range (USA, near Lake Superior):** Iron ore.
- **Karaganda (Kazakhstan) & Donbas (Ukraine):** Coal.
- **Postmasburg (S. Africa):** Manganese.

FACTS